

Nonreal Roots Survive in Polynomial Iterative Equations

A cubic counterexample and a classification of circular spectra

Part II: the increasing spectral core on several circles

Research draft prepared for Vladimir Reshetnikov

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Abstract

For a real polynomial $P(z) = \sum_{j=0}^d a_j z^j$, write $P[f](x) = \sum_{j=0}^d a_j f^{\circ j}(x)$, where $f^{\circ 0}(x) = x$. A published question of Draga and Morawiec asks whether nonreal characteristic roots of maximal modulus can be removed when a real root has the same modulus. We give a negative answer under the stated continuity assumptions. An explicit, odd, increasing, globally bi-Lipschitz homeomorphism F satisfies $F^{\circ 3}(x) = 8x$, although $F(1) = 3$. Its minimal iterative polynomial is $z^3 - 8$, certified by a 3×3 Hankel determinant equal to -56 . A second construction is real analytic at every real point and has minimal polynomial $(z - 1)^2(z^2 + 1)$; deleting its nonreal roots also fails.

We then determine exactly which monic real polynomials supported on one circle $|z| = r > 0$ occur as minimal iterative polynomials of increasing homeomorphisms of $\mathbb{R}$. For $r \neq 1$, the positive root r must have odd multiplicity, at least as large as every other multiplicity. For $r = 1$, the identity is exceptional; otherwise the multiplicity of 1 must be even and strictly larger than every other multiplicity. Every permitted polynomial is realized by an explicit small trigonometric perturbation of a polynomial-exponential coordinate. The proof uses two-sided orbit recurrences and an elementary positivity lemma. We also obtain sharp increasing-solution rigidity criteria and explain the role of differentiability at a fixed point. Exact rational verification code, certificates, and the random area-selection record accompany the article.

Status. The counterexamples and classification are proved below. This is an AI-assisted, unrefereed research draft, not a proof-assistant-checked publication. A targeted literature search located the published open question but did not establish priority for the answer or for the classification. The conjugacy mechanism itself is not claimed to be new.

Part II (added 29 September 2026, batch 41; from Section 12 on) removes the single-circle restriction for increasing homeomorphisms of $\mathbb{R}$: it classifies the minimal iterative polynomials with arbitrary root moduli (Theorem 12.2), gives the exact universal order reduction $\mathcal{R}(P)$ of every equation with nonzero constant coefficient (Theorem 16.3), and decides affine rigidity and the existence of fixed points. It is likewise an AI-assisted, unrefereed draft with no priority claim.

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1 The problem selected and the scope of the answer

1.1 A precise published target

Let $I \subseteq \mathbb{R}$ be a nondegenerate interval and let $f : I \rightarrow I$ be continuous. An equation

$$a_d f^{\circ d}(x) + a_{d-1} f^{\circ d-1}(x) + \cdots + a_1 f(x) + a_0 x = 0 \quad (1)$$

with $a_0 a_d \neq 0$ is a polynomial-like iterative equation. Its *characteristic polynomial* is $P(z) = \sum_{j=0}^d a_j z^j$. The notation concerns composition, not pointwise powers: in general, $f^{\circ 2}(x)$ is not $f(x)^2$.

The specific target is Problem 6.1 of Draga and Morawiec [1], in the last section of their paper. In the context of lowering the order of (1), they ask about deleting nonreal roots of largest modulus when a real root has that same modulus. The question also appears in the authors' preprint, page 13. Here *deleting roots* means replacing P by the corresponding polynomial divisor while preserving the implication that every solution satisfies the lower-order equation.

The answer under these hypotheses alone is negative. In fact, the cubic counterexample in Section 2 satisfies no nonzero equation of lower order whatsoever. The analytic counterexample in Section 3 shows that lack of smoothness is not the general reason for failure.

The equal-modulus qualification is essential. Existing order-reduction results employ strict separation of root moduli, together with orbit monotonicity or alternating monotonicity; Draga's later note discusses additional elimination cases and again identifies the unrestricted nonreal-root issue as unresolved [2]. Our examples do not contradict those theorems: they lie at the spectral boundary excluded by strict separation.

1.2 Relation to the initial candidate

The literature search began with Petrov's MathOverflow question asking for conditions on P that force a continuous solution of $P[f] = 0$ to be linear [3]. The retrieved question had no posted answer. It led to the more sharply formulated root-elimination problem above.

Throughout this article, *homogeneous linear* means $f(x) = cx$ and *affine* means $f(x) = cx + b$. The distinction matters when 1 is a multiple root. We do not claim a complete answer to Petrov's question for arbitrary polynomials or for both monotonicity orientations. We do obtain an exact answer for *increasing solutions when all characteristic roots lie on one circle*; see Section 8. [Added 29 September 2026, batch 41: Part II extends this answer to increasing solutions with arbitrary root moduli, Theorem 17.1.]

1.3 Random selection, exclusion, and priority

Before choosing a problem, a list of 80 mathematical areas was supplied to an operating-system-backed random generator through `secrets.randbelow(80)`. The first draw returned the zero-based index 17: area 18, *Functional equations*. There was no redraw. The complete ordered list and the result are preserved in `data/selection.json`.

The supplied `manifest(1).tex` was reviewed in full. The present problem was not selected from its entries. Its SHA-256 fingerprint is recorded with the selection so that the exclusion check can be tied to the actual supplied version; see Appendix B.

The target is therefore an explicitly published open question, not a new question invented for this report. The search found the original statement, subsequent related papers, and no explicit later answer. That is not

an exhaustive bibliographic certification that the question remained open up to the preparation date. The mathematical conclusion below is independent of that priority issue: the proposed general elimination is false as printed. Nor should the elementary conjugacy construction be advertised as a new technique. The contribution developed here is the explicit counterexample, its connection with the published question, and the self-contained spectral classification.

2 An exact cubic counterexample

2.1 The two-branch definition

Define a function on one fundamental interval by

$$\phi(u) = \begin{cases} (u+5)/2, & 1 \leq u \leq 3, \\ 4u-8, & 3 \leq u \leq 8. \end{cases} \quad (2)$$

The formulas agree at $u = 3$. For $x > 0$, choose the unique $k \in \mathbb{Z}$ such that $8^k \leq x < 8^{k+1}$, and put

$$F(x) = 8^k \phi(8^{-k}x). \quad (3)$$

Complete the definition by

$$F(0) = 0, \quad F(-x) = -F(x) \quad (x > 0). \quad (4)$$

The endpoint identity $\phi(8) = 24 = 8\phi(1)$ makes the scaling construction consistent. It gives, on the whole real line,

$$F(8x) = 8F(x). \quad (5)$$

Theorem 2.1 (Cubic counterexample). *The function F defined by (2)–(4) is an odd, strictly increasing homeomorphism of $\mathbb{R}$ with*

$$\frac{1}{2}|x-y| \leq |F(x) - F(y)| \leq 4|x-y| \quad (x, y \in \mathbb{R}). \quad (6)$$

It satisfies

$$F^{\circ 3}(x) = 8x \quad (x \in \mathbb{R}), \quad (7)$$

but is not homogeneous linear or affine. The smallest degree of a nonzero polynomial Q with $Q[F] = 0$ is three, and the monic polynomial of that degree is

$$\mu_F(z) = z^3 - 8 = (z-2)(z^2 + 2z + 4). \quad (8)$$

Proof. On every scaled copy of the two intervals in (2), the slope of F is either $1/2$ or 4 . The pieces meet continuously. Moreover,

$$\frac{4}{3} \leq \frac{F(x)}{x} \leq 3 \quad (x > 0). \quad (9)$$

For the first branch this follows from $\phi(u)/u = 1/2 + 5/(2u)$; for the second it follows from $\phi(u)/u = 4 - 8/u$. Thus $F(x) \rightarrow 0$ as $x \rightarrow 0$, and $F(x) \rightarrow \infty$ as $x \rightarrow \infty$. Oddness handles the negative half-line. Strict monotonicity and surjectivity follow.

Between two strictly positive points there are only finitely many breakpoints, so their secant slope lies between $1/2$ and 4 . Pass to a limit when one endpoint is zero, and use oddness and addition across zero to obtain (6) for all pairs of points.

To check the iterate relation it suffices to work on $[1, 8]$. There are three relevant itinerary intervals, because an intermediate iterate crosses a breakpoint at $x = 4$. Direct substitution gives Table 1. In using the table, values in $[8, 64]$ are computed with (5). The final column is $8x$ in every row. Scaling extends the identity to every positive x ; oddness and the value at zero finish the proof of (7).

The values $F(1) = 3$ and $F(3) = 4$ show that F is not homogeneous linear. Since $F(0) = 0$, it is not affine either. Minimality is proved immediately below. $\square$

Table 1: Exact branchwise verification of the cubic relation.

Initial interval	$F(x)$	$F^{\circ 2}(x)$	$F^{\circ 3}(x)$
$[1, 3]$	$(x + 5)/2$	$2x + 2$	$8x$
$[3, 4]$	$4x - 8$	$16x - 40$	$8x$
$[4, 8]$	$4x - 8$	$2x + 16$	$8x$

2.2 A small exact certificate of minimality

The orbit beginning at 1 is

$$1, 3, 4, 8, 24, 32, 64, 192, 256, \dots \quad (10)$$

In particular,

$$\det \begin{pmatrix} 1 & 3 & 4 \\ 3 & 4 & 8 \\ 4 & 8 & 24 \end{pmatrix} = -56 \neq 0. \quad (11)$$

Suppose $q_0x + q_1F(x) + q_2F^{\circ 2}(x) = 0$ on $\mathbb{R}$. Substitution of $x = 1, F(1), F^{\circ 2}(1)$ gives the matrix in (11) times $(q_0, q_1, q_2)^T$ equal to zero. Hence all three coefficients vanish. This rules out every nonzero relation of degree at most two. Relation (7) supplies a monic relation of degree three. Subtracting any other monic cubic relation would produce a relation of lower degree, so (8) is unique.

The roots of μ_F are

$$2, \quad -1 + i\sqrt{3}, \quad -1 - i\sqrt{3}. \quad (12)$$

They all have modulus 2. Deleting the nonreal pair leaves the polynomial $z - 2$, but $(z - 2)[F](1) = 1 \neq 0$. This is precisely the obstruction asked about in the selected problem. Degree three is also the smallest possible degree for a real polynomial to have both a real root and a nonreal conjugate pair.

2.3 The conjugacy behind the formula

The construction can also be understood without the itinerary table. On $[1, 8]$ define

$$h_0(u) = \begin{cases} 2u - 1, & 1 \leq u \leq 2, \\ u/2 + 2, & 2 \leq u \leq 4, \\ u, & 4 \leq u \leq 8. \end{cases} \quad (13)$$

Extend it by $H(8^k u) = 8^k h_0(u)$ for $u \in [1, 8]$, then extend oddly and set $H(0) = 0$. The map H is an increasing homeomorphism, and it commutes with $D_8(x) = 8x$. Direct calculation shows

$$F = H \circ D_2 \circ H^{-1}. \quad (14)$$

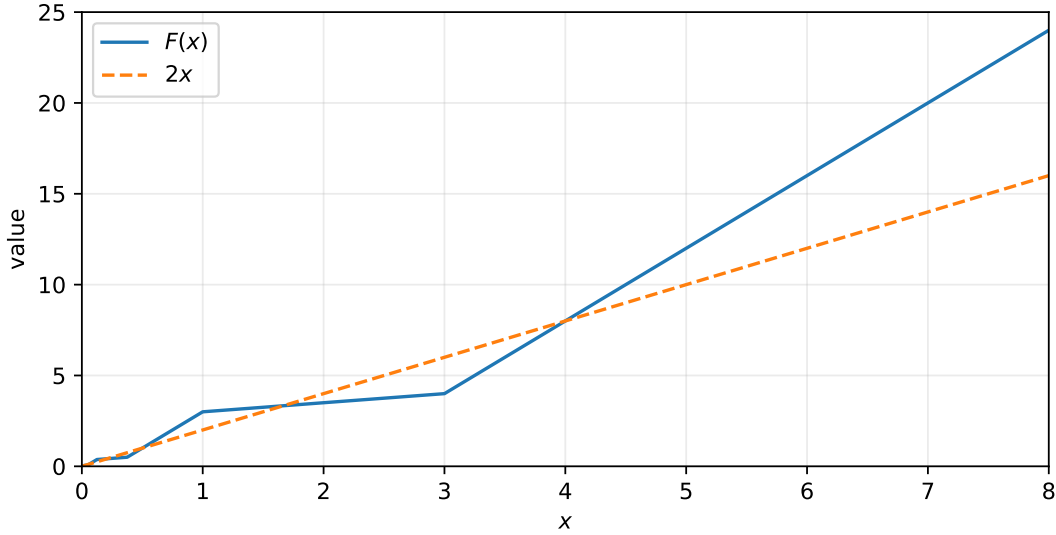


Figure 1: The exact cubic counterexample on one scale interval, compared with the homogeneous linear solution. There are infinitely many scaled copies of the two affine branches, accumulating at zero.

Consequently,

$$F^{\circ 3} = H \circ D_8 \circ H^{-1} = D_8. \quad (15)$$

The middle equality is useful only because H commutes with D_8 ; it need not commute with D_2 .

There is a general warning here. Arbitrary nonlinear conjugacy does *not* preserve an arbitrary linear combination of iterates. Although iterates conjugate one by one, a nonlinear conjugating map cannot be moved through a sum. In (14), commutation with the specific third iterate supplies the extra fact needed. In later constructions, we impose a linear difference equation on the conjugating coordinate itself instead.

2.4 A visible failure of differentiability

The scaled sequences $x_n = 8^{-n}$ and $y_n = 3 \cdot 8^{-n}$ both tend to zero, but

$$\frac{F(x_n)}{x_n} = 3, \quad \frac{F(y_n)}{y_n} = \frac{4}{3}. \quad (16)$$

Thus F is not differentiable at zero, despite being globally bi-Lipschitz. Section 9 explains why this is inevitable for a square-free circular spectrum of radius different from 1. The next example avoids this regularity issue altogether.

3 A real-analytic counterexample on the entire real line

Let

$$h(t) = t + \frac{1}{4} \sin \frac{\pi t}{2}, \quad f(x) = h(h^{-1}(x) + 1). \quad (17)$$

Here h^{-1} is the ordinary inverse function, not a reciprocal. Since

$$h'(t) = 1 + \frac{\pi}{8} \cos \frac{\pi t}{2} \quad \text{and} \quad 1 - \frac{\pi}{8} > 0, \quad (18)$$

h is strictly increasing. Also $|h(t) - t| \leq 1/4$, so it maps $\mathbb{R}$ onto $\mathbb{R}$. Its derivative never vanishes. The real-analytic inverse function theorem, applied at each real point, makes h^{-1} and f real analytic. No global complex-analytic inverse is asserted or required.

Theorem 3.1 (Analytic quartic counterexample). *The function in (17) is an increasing, globally bi-Lipschitz, real-analytic homeomorphism of $\mathbb{R}$. Its minimal iterative polynomial is*

$$\mu_f(z) = (z - 1)^2(z^2 + 1) = z^4 - 2z^3 + 2z^2 - 2z + 1. \quad (19)$$

It also satisfies $f^{\circ 4}(x) = x + 4$. The equation obtained by deleting the nonreal roots from (19) is not satisfied.

Proof. Conjugacy gives, for every integer n ,

$$f^{\circ n}(h(t)) = h(t + n) = t + n + \frac{1}{4} \sin \frac{\pi(t + n)}{2}. \quad (20)$$

The linear polynomial in n is killed by $(E - 1)^2$, where E shifts n to $n + 1$. The sine term is killed by $E^2 + 1$. The two operators commute, so their product kills the entire sequence. Therefore $\mu_f[f] = 0$. Periodicity of the sine also gives $h(t + 4) = h(t) + 4$, proving $f^{\circ 4}(x) = x + 4$.

The first orbit values at zero are

$$0, \quad \frac{5}{4}, \quad 2, \quad \frac{11}{4}, \quad 4, \quad \frac{21}{4}, \quad 6, \dots \quad (21)$$

Their 4×4 Hankel determinant is

$$\det \begin{pmatrix} 0 & 5/4 & 2 & 11/4 \\ 5/4 & 2 & 11/4 & 4 \\ 2 & 11/4 & 4 & 21/4 \\ 11/4 & 4 & 21/4 & 6 \end{pmatrix} = 1. \quad (22)$$

The same argument used for (11) rules out every relation of degree below four. This proves minimality.

After deleting i and $-i$, the remaining polynomial is $(z - 1)^2$. At $x = 0$ its residual is

$$f^{\circ 2}(0) - 2f(0) + 0 = 2 - 2 \cdot \frac{5}{4} = -\frac{1}{2}. \quad (23)$$

Thus the proposed deletion fails. Finally, $f'(h(t)) = h'(t + 1)/h'(t)$ is bounded above and bounded away from zero, which proves the bi-Lipschitz assertion. $\square$

All roots in (19) lie on the unit circle, and 1 is a real root of maximal modulus. This gives a second negative answer, now with real analyticity at every real point and with all polynomial coefficients nonzero. Notice that the map has no fixed point: $h(t + 1) > h(t)$. The distinction between a fixed point and a translation-like orbit will account for the different multiplicity rules at radius 1.

4 Orbit recurrences and minimal iterative polynomials

We now develop the tools for a general classification. The results in this section also explain why an exact orbit determinant is a valid minimality certificate.

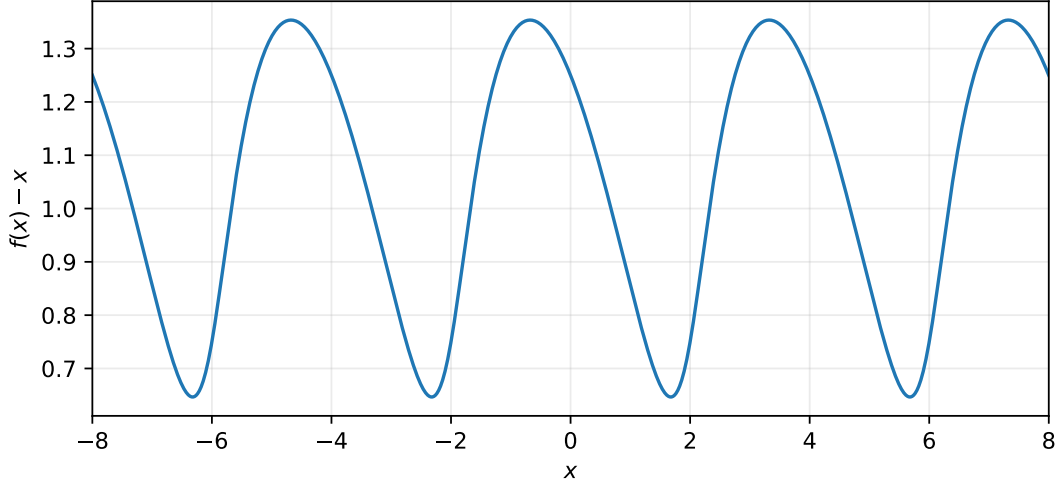


Figure 2: Numerical illustration of $f(x) - x$ for (17). The displacement is nonconstant and 4-periodic. The identities and the counterexample are established exactly in the text, not by this plot.

4.1 The annihilator ideal

Definition 4.1. For a function $f : \mathbb{R} \rightarrow \mathbb{R}$, let

$$\text{Ann}(f) = \{P \in \mathbb{R}[z] : P[f](x) = 0 \text{ for every } x \in \mathbb{R}\}. \quad (24)$$

If this set contains a nonzero polynomial, its monic polynomial of least degree is called the *minimal iterative polynomial* and is denoted by μ_f .

Lemma 4.2. *The set $\text{Ann}(f)$ is an ideal of $\mathbb{R}[z]$. When it is nonzero, $\text{Ann}(f) = (\mu_f)$.*

Proof. Closure under addition and scalar multiplication is immediate. For every polynomial P ,

$$(zP)[f](x) = P[f](f(x)). \quad (25)$$

Thus $P \in \text{Ann}(f)$ implies $zP \in \text{Ann}(f)$, and consequently every polynomial multiple of P belongs to the ideal. Divide any annihilator by the monic least-degree one. The remainder is another annihilator and has smaller degree, so it must be zero. $\square$

It is *right composition* in (25) that respects linear combinations. Applying f to a linear combination of its iterates would not give the same conclusion.

4.2 Why both forward and backward iterates exist

Lemma 4.3. *If $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous and $P[f] = 0$ for a polynomial with $P(0) \neq 0$, then f is a strictly monotone homeomorphism of $\mathbb{R}$.*

Proof. If $f(x) = f(y)$, then every positive iterate has the same value at x and y . Subtracting the two equations $P[f](x) = P[f](y) = 0$ gives $P(0)(x - y) = 0$, hence $x = y$. A continuous injective real function is strictly monotone.

If its image were a proper interval, at least one of its limits at $+\infty$ and $-\infty$ would be finite. Along the corresponding end, $f(x)$ would tend to a finite number, and continuity would make every positive iterate tend to a finite number as well. But the term $P(0)x$ in the equation would be unbounded. This contradiction proves surjectivity. A continuous strictly monotone bijection of $\mathbb{R}$ has a continuous inverse. $\square$

For such an f , define f^{on} for every $n \in \mathbb{Z}$ using inverse iterates when $n < 0$. For any starting point x , its orbit $u_x(n) = f^{on}(x)$ satisfies

$$\sum_{j=0}^d a_j u_x(n+j) = 0 \quad (n \in \mathbb{Z}). \quad (26)$$

Indeed, substitute $f^{on}(x)$ into $P[f] = 0$.

4.3 Exponential-polynomial solutions

Lemma 4.4 (Two-sided recurrence decomposition). *Let P be a polynomial with nonzero leading and constant coefficients. Write its distinct complex roots as λ , with multiplicities m_λ . Every complex sequence $u : \mathbb{Z} \rightarrow \mathbb{C}$ satisfying $P(E)u = 0$ has a unique representation*

$$u(n) = \sum_{\lambda} p_{\lambda}(n) \lambda^n, \quad \deg p_{\lambda} < m_{\lambda}. \quad (27)$$

For a real sequence, conjugate roots have conjugate coefficient polynomials. Its minimal recurrence polynomial is

$$\prod_{p_{\lambda} \neq 0} (z - \lambda)^{1 + \deg p_{\lambda}}. \quad (28)$$

Proof. For a polynomial p ,

$$(E - \mu)(p(n)\lambda^n) = \lambda^n(\lambda p(n+1) - \mu p(n)). \quad (29)$$

When $\mu = \lambda$, a nonconstant polynomial degree drops by one, and a constant is killed. When $\mu \neq \lambda$, the degree is preserved and the leading coefficient is multiplied by $\lambda - \mu$.

These observations prove that the proposed sequences satisfy the recurrence. They also prove their linear independence: in any putative linear relation, apply enough factors corresponding to all roots except one. The surviving root component is a nonzero polynomial times an exponential unless its coefficient polynomial was zero. There are $\deg P$ independent sequences. A two-sided recurrence with nonzero first and last coefficients is determined by $\deg P$ consecutive values, so these sequences form a basis. Formula (29) then proves the exact multiplicities in (28). $\square$

Lemma 4.5 (From individual orbits to the whole function). *The minimal iterative polynomial is the least common multiple of the minimal recurrence polynomials of all scalar orbits u_x . In particular, for each root, its multiplicity in μ_f is the maximum of the multiplicities with which it occurs in the scalar orbits.*

Proof. Each orbit is annihilated by μ_f , so its minimal recurrence polynomial divides μ_f . Conversely, any common multiple of all the orbit polynomials annihilates every orbit at $n = 0$, and hence annihilates f . Only finitely many roots and bounded multiplicities are possible, because they all occur in μ_f . Therefore their least common multiple and each stated maximum are attained using finitely many orbits. $\square$

This also gives a finite-dimensional interpretation. Right composition by f acts linearly on the span of $\text{id}, f, f^{\circ 2}, \dots$. The vector id is cyclic, and μ_f is the minimal polynomial on this cyclic space. The function f itself need not be linear as a map of real numbers.

5 The positivity lemma that controls a circular spectrum

A finite exponential polynomial on the unit circle is a sequence

$$v(n) = \sum_{\zeta \in S} p_{\zeta}(n) \zeta^n, \quad n \in \mathbb{Z}, \quad (30)$$

where S is a finite set of distinct complex numbers of modulus one. The sequence is real when the terms respect conjugation. The root $\zeta = 1$ contributes the nonoscillating polynomial part.

5.1 A recurrence fact for finite trigonometric sequences

Lemma 5.1. *For any finite set S on the unit circle, there are positive integers $N_k \rightarrow \infty$ such that $\zeta^{N_k} \rightarrow 1$ simultaneously for every $\zeta \in S$. Consequently, for a finite trigonometric sequence $T(n) = \sum_{\zeta \in S} c_{\zeta} \zeta^n$ and any fixed $\ell \in \mathbb{Z}$,*

$$T(N_k + \ell) \rightarrow T(\ell), \quad T(-N_k + \ell) \rightarrow T(\ell). \quad (31)$$

Proof. The vectors $(\zeta^n)_{\zeta \in S}$ lie in a compact finite product of circles. Choose a convergent subsequence with indices n_k whose successive gaps tend to infinity. The componentwise quotient of the vectors at indices n_{2k} and n_k tends to the identity vector. Set $N_k = n_{2k} - n_k$. The inverse quotients converge to the identity as well, giving both assertions in (31). $\square$

Lemma 5.2 (Nonnegative trigonometric sequences have positive mean). *Let $T(n) = \sum_{\zeta \in S} c_{\zeta} \zeta^n$ be real, nonnegative for every $n \in \mathbb{Z}$, and not identically zero. Then $1 \in S$ and $c_1 > 0$.*

Proof. The Cesàro mean of ζ^n over $n = 0, \dots, N-1$ tends to zero for $\zeta \neq 1$, by the finite geometric-sum formula. Thus the mean of T tends to c_1 , interpreted as zero if $1 \notin S$. Distinctness of the frequencies gives

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} |T(n)|^2 = \sum_{\zeta \in S} |c_{\zeta}|^2 > 0. \quad (32)$$

On the other hand, $0 \leq T(n) \leq M$ for some finite M , and $T(n)^2 \leq MT(n)$. Taking means proves $c_1 > 0$. $\square$

5.2 Parity and dominance of the nonoscillating term

Theorem 5.3 (Two-sided positivity). *Suppose the real sequence (30) is nonnegative for all $n \in \mathbb{Z}$ and is not identically zero. Let $d = \max\{\deg p_{\zeta} : p_{\zeta} \neq 0\}$. Then d is even. The polynomial p_1 is present, has degree exactly d , and has positive leading coefficient. The analogous statement for a nonpositive sequence holds after multiplication by -1 .*

Proof. Let c_{ζ} be the coefficient of n^d in p_{ζ} , with zero used when its degree is lower, and set

$$T(n) = \sum_{\zeta \in S} c_{\zeta} \zeta^n. \quad (33)$$

The leading trigonometric sequence is not identically zero, by the independence in Lemma 4.4. Uniformly as $|n| \rightarrow \infty$,

$$v(n) = n^d T(n) + O(|n|^{d-1}), \quad (34)$$

with the evident exact interpretation when $d = 0$.

At $+\infty$, nonnegativity gives $\liminf T(n) \geq 0$. Use the first sequence in (31) to conclude $T(\ell) \geq 0$ for every fixed integer ℓ . At $-\infty$, division by $|n|^d$ and the second return sequence give $(-1)^d T(\ell) \geq 0$ for every ℓ . If d were odd, T would vanish identically, a contradiction. Thus d is even. Lemma 5.2 now gives $c_1 > 0$. Hence p_1 has degree d and the asserted leading sign. $\square$

The use of both time directions is decisive. For a recurrence constrained only at nonnegative indices, an odd-degree polynomial may remain positive. A homeomorphism supplies all negative iterates, eliminating that possibility.

Corollary 5.4 (A vanishing unit-circle exponential polynomial). *If an exponential polynomial (30) tends to zero as $n \rightarrow +\infty$ or as $n \rightarrow -\infty$, it is identically zero.*

Proof. Choose its largest polynomial degree d and its leading trigonometric sequence T as in (34). The assumed limit implies $T(n) \rightarrow 0$ in the chosen direction. The corresponding return sequence in Lemma 5.1 gives $T(\ell) = 0$ for every ℓ , contrary to the choice of a nonzero leading degree. Thus a nonzero exponential polynomial cannot have the stated limit. $\square$

6 Classification of minimal polynomials on one circle

Call a polynomial *circular of radius r* if every one of its complex roots has modulus r . Multiplicities are counted in the usual algebraic sense. A missing root is assigned multiplicity zero.

Theorem 6.1 (Circular-spectrum classification). *Let $Q \in \mathbb{R}[z]$ be nonconstant and monic, and suppose all its roots lie on $|z| = r$ for some $r > 0$. Then Q is the minimal iterative polynomial of an increasing homeomorphism $f : \mathbb{R} \rightarrow \mathbb{R}$ if and only if the following conditions hold.*

- (i) *If $r \neq 1$, the positive real number r is a root of odd multiplicity m_r , and every other root has multiplicity at most m_r .*
- (ii) *If $r = 1$, either $Q(z) = z - 1$, or 1 is a root of even multiplicity $m_1 \geq 2$ and every other root has multiplicity at most $m_1 - 1$.*

In case (i), realizations can be chosen odd, globally bi-Lipschitz, and real analytic away from zero. In the nonidentity part of case (ii), realizations can be chosen globally bi-Lipschitz and real analytic at every real point; they have no fixed point.

This is a classification of the polynomials that can be *minimal*. It is not a parametrization of every function realizing any one polynomial. The minimality qualification is essential: for example, the identity satisfies $(z - 1)^7 [\text{id}] = 0$, but its minimal polynomial is only $z - 1$. [Added 29 September 2026, batch 41: Theorem 12.2 of Part II classifies the minimal polynomials with roots of arbitrary moduli; on one circle it reduces to the present theorem (Remark 15.3).]

6.1 Necessity when $r \neq 1$

Let f be increasing and let $\mu_f = Q$. By Lemma 4.4, every orbit has the form

$$u_x(n) = r^n \sum_{|\zeta|=1} p_{x,\zeta}(n) \zeta^n. \quad (35)$$

If $r > 1$, this tends to zero as $n \rightarrow -\infty$; if $r < 1$, it tends to zero as $n \rightarrow +\infty$. Passing to this limit in $f(u_x(n)) = u_x(n+1)$ gives $f(0) = 0$. Since f and f^{-1} are increasing and fix zero, every nonzero orbit keeps the sign of its starting point for all $n \in \mathbb{Z}$.

For $x > 0$, divide (35) by $r^n > 0$ and apply Theorem 5.3. Its largest polynomial degree is even and is attained by the coefficient of $\zeta = 1$, corresponding to the characteristic root r . The same conclusion holds for $x < 0$ after a sign change. Thus, in every nonzero scalar orbit, the multiplicity of r is odd and is at least as large as every other multiplicity.

Now take the maxima over orbits using Lemma 4.5. The maximum of a nonempty finite set of odd multiplicities is odd. Every other global multiplicity is bounded by that maximum. This proves the necessity of condition (i).

6.2 Necessity on the unit circle

Suppose $r = 1$ and $f \neq \text{id}$. Choose a nonfixed starting point x . Because f is increasing, the difference sequence

$$v_x(n) = u_x(n+1) - u_x(n) \quad (36)$$

has a strict, constant sign for all $n \in \mathbb{Z}$.

Taking a first difference lowers the degree of the polynomial component at the root 1 by one. At every other root it preserves the degree, as (29) shows. Apply Theorem 5.3 to v_x , after a sign change if necessary. Its largest degree is even and is attained at 1. Therefore the polynomial component of u_x at 1 has odd degree $d \geq 1$, and every oscillatory component has degree at most $d - 1$. Equivalently, this orbit has an even multiplicity $d + 1$ at 1, strictly larger than all the other multiplicities.

The nonoscillating leading term dominates the orbit at both ends:

$$u_x(n) = an^d + O(|n|^{d-1}), \quad a \neq 0, \quad d \text{ odd}. \quad (37)$$

In particular, the two time tails go to opposite infinities. If f had a fixed point, an increasing orbit starting on either side of it could never cross it, contradicting (37). Hence a nonidentity f in this setting has no fixed point, and all its orbits are nonfixed. Lemma 4.5 now gives an even maximal multiplicity at 1 and the strict bound on every other multiplicity. If instead $f = \text{id}$, the minimal polynomial is $z - 1$. This proves condition (ii).

6.3 Encoding the remaining root directions

For the construction, write every root other than the distinguished positive root using an angle $0 < \theta_j \leq \pi$. An angle $0 < \theta_j < \pi$ denotes the conjugate pair $re^{i\theta_j}, re^{-i\theta_j}$, with common multiplicity m_j . The angle $\theta_j = \pi$ denotes the single real root $-r$, if present, with multiplicity m_j . Choose distinct angles and set

$$e_j = m_j - 1. \quad (38)$$

A term $r^{e_j} \cos(\theta_j t)$ then produces precisely the desired degree in its sampled orbit. At angle π , the two exponential expressions for the cosine coincide on integer samples and give the one real root, not two roots.

6.4 Construction for $r \neq 1$

Write $m_r = 2k + 1$ and set $a = \log r \neq 0$. Choose $C \geq 1$ so large that

$$\frac{k}{\sqrt{C}} \leq \frac{|a|}{2}, \quad (39)$$

and define

$$A(t) = (t^2 + C)^k, \quad W(t) = A(t) + \sum_j \varepsilon_j t^{e_j} \cos(\theta_j t), \quad h(t) = r^t W(t), \quad (40)$$

where every ε_j is positive and sufficiently small. The permitted multiplicities give $0 \leq e_j \leq 2k$. An explicit sufficient choice is any collection satisfying

$$\sum_j \varepsilon_j (|a| + e_j + \theta_j) \leq \frac{|a|}{4}. \quad (41)$$

When there are no other roots, the sum is empty.

Here are the estimates that justify the construction. For $0 \leq e \leq 2k$,

$$|t|^e \leq A(t), \quad e|t|^{e-1} \leq eA(t) \quad (e \geq 1), \quad \frac{|A'(t)|}{A(t)} \leq \frac{k}{\sqrt{C}}. \quad (42)$$

The first inequalities follow separately on $|t| \leq 1$ and $|t| \geq 1$. The last one follows by maximizing $2k|t|/(t^2 + C)$. Since (41) implies $\sum_j \varepsilon_j \leq 1/4$,

$$\frac{3}{4}A(t) \leq W(t) \leq \frac{5}{4}A(t). \quad (43)$$

Also

$$h'(t) = r^t (aW(t) + W'(t)). \quad (44)$$

The unperturbed expression $aA + A'$ has the sign of a and absolute value at least $|a|A/2$. Each perturbing expression

$$at^{e_j} \cos(\theta_j t) + e_j t^{e_j-1} \cos(\theta_j t) - \theta_j t^{e_j} \sin(\theta_j t) \quad (45)$$

has absolute value at most $(|a| + e_j + \theta_j)A(t)$, with the middle term omitted for $e_j = 0$. Thus

$$\operatorname{sgn} h'(t) = \operatorname{sgn} a, \quad \frac{|a|}{4} r^t A(t) \leq |h'(t)| \leq \frac{7|a|}{4} r^t A(t). \quad (46)$$

It follows that h is a real-analytic bijection from $\mathbb{R}$ onto $(0, \infty)$: it is increasing for $r > 1$ and decreasing for $r < 1$. Define

$$f(x) = h(h^{-1}(x) + 1) \quad (x > 0), \quad f(0) = 0, \quad f(-x) = -f(x). \quad (47)$$

Even when h is decreasing, the composition in (47) is increasing. Its limit at zero is zero, so the odd extension is an increasing homeomorphism of $\mathbb{R}$.

Every orbit on the positive half-line is $h(t + n)$. Expanding its cosine terms shows that it is annihilated by Q . Oddness handles negative orbits. More importantly, at the starting point $h(0)$ the sampled sequence is

$$h(n) = r^n \left((n^2 + C)^k + \sum_j \varepsilon_j n^{e_j} \cos(\theta_j n) \right). \quad (48)$$

The component at r has degree exactly $2k$, and every other component has exactly its assigned degree e_j , with nonzero leading coefficient. Lemma 4.4 proves that this single orbit has minimal polynomial Q . Hence $\mu_f = Q$.

Finally, (46) bounds $f'(h(t)) = h'(t+1)/h'(t)$ above and away from zero, because $A(t+1)/A(t)$ is positive, continuous, and tends to 1 at both ends. The resulting derivative bounds extend to secant bounds across zero. This proves the bi-Lipschitz and regularity assertions in case (i).

6.5 Construction on the unit circle

For the identity case choose $f = \text{id}$. Otherwise write $m_1 = 2k + 2$ and put

$$A(t) = \int_0^t (1 + s^2)^k ds = \sum_{\ell=0}^k \binom{k}{\ell} \frac{t^{2\ell+1}}{2\ell+1}. \quad (49)$$

Now A is an odd polynomial of degree $2k + 1$ and $A'(t) = (1 + t^2)^k > 0$. Since $e_j \leq 2k$, set

$$h(t) = A(t) + \sum_j \varepsilon_j t^{e_j} \cos(\theta_j t) \quad (50)$$

with all $\varepsilon_j > 0$ chosen so that

$$\sum_j \varepsilon_j (e_j + \theta_j) < \frac{1}{2}. \quad (51)$$

The same estimates as in (42), now relative to A' , imply

$$\frac{1}{2}(1 + t^2)^k < h'(t) < \frac{3}{2}(1 + t^2)^k. \quad (52)$$

The polynomial A has degree one more than every perturbing amplitude, so $h(t) \rightarrow \pm\infty$ as $t \rightarrow \pm\infty$. Therefore h is a real-analytic increasing diffeomorphism of $\mathbb{R}$.

Set $f = h \circ T \circ h^{-1}$, where $T(t) = t + 1$. All its orbits are $h(t + n)$ and are annihilated by Q . The orbit $h(n)$ contains a nonoscillating polynomial of degree $2k + 1$ and exactly the prescribed oscillating polynomial degrees e_j . Thus its minimal recurrence polynomial, and hence μ_f , is Q . The derivative ratio from (52) is bounded above and away from zero, proving global bi-Lipschitz continuity. The conjugacy also gives $f(x) > x$ everywhere and therefore no fixed point. This completes the sufficiency proof and the proof of Theorem 6.1.

7 Quantitative square-free realizations

The simplest part of the classification has a particularly transparent formula. It shows that survival of nonreal roots is not restricted to roots of unity or to an isolated cubic.

Fix $r > 0$, $r \neq 1$, and distinct angles $0 < \theta_j \leq \pi$. Choose nonzero real amplitudes ε_j and put

$$W(t) = 1 + \sum_{j=1}^q \varepsilon_j \cos(\theta_j t), \quad h(t) = r^t W(t). \quad (53)$$

Let $a = \log r$ and

$$B = \sum_{j=1}^q |\varepsilon_j| \sqrt{a^2 + \theta_j^2}. \quad (54)$$

Proposition 7.1. *If $B < |a|$, then the odd extension of $f = h \circ T \circ h^{-1}$ from $(0, \infty)$ to $\mathbb{R}$ is an increasing homeomorphism with minimal polynomial*

$$(z - r) \prod_{0 < \theta_j < \pi} (z^2 - 2r \cos \theta_j z + r^2) \prod_{\theta_j = \pi} (z + r). \quad (55)$$

It is real analytic away from zero, and its positive derivative on either open half-line satisfies

$$r \frac{|a| - B}{|a| + B} \leq f'(x) \leq r \frac{|a| + B}{|a| - B}. \quad (56)$$

These are also valid global lower and upper Lipschitz constants.

Proof. Since $B \geq |a| \sum_j |\varepsilon_j|$, the condition gives $W(t) > 0$. Moreover,

$$aW(t) + W'(t) = a + \sum_j \varepsilon_j (a \cos(\theta_j t) - \theta_j \sin(\theta_j t)) \quad (57)$$

differs from a by at most B . Thus h' has the sign of a , and its absolute value lies between $r^t(|a| - B)$ and $r^t(|a| + B)$. This proves that h is a suitable coordinate, and taking the ratio at $t + 1$ and t gives (56). Minimality follows from the nonzero frequency components in $h(n)$, exactly as before. $\square$

For example, taking one angle $0 < \theta < \pi$ yields a nonlinear solution whose minimal polynomial is

$$(z - r)(z^2 - 2r \cos \theta z + r^2). \quad (58)$$

The angle can be any such real number, not necessarily a rational multiple of π . In particular, $\theta = \pi/2$ and $r = 2$ give the cubic $z^3 - 2z^2 + 4z - 8$, all of whose coefficients are nonzero.

A small oscillatory component can coexist with strict monotonicity because the exponential envelope supplies a nonzero derivative margin. It is incorrect to infer that an orbit must lose every oscillatory mode merely because the unnormalized orbit is monotone. The cubic orbit (10) makes this visible:

$$\frac{F^{on}(1)}{2^n} = 1, \frac{3}{2}, 1, 1, \frac{3}{2}, 1, \dots \quad (59)$$

The normalized values oscillate, while the actual orbit is strictly increasing.

8 Sharp rigidity consequences for increasing solutions

The classification concerns minimal polynomials, but a given equation need not be minimal. Passing to divisors gives explicit rigidity tests.

Corollary 8.1 (Nonunit circle). *Let P be a nonconstant monic real polynomial whose roots lie on $|z| = r \neq 1$. If r is not a root, there is no increasing continuous solution of $P[f] = 0$ on $\mathbb{R}$. If r is a root, every increasing continuous solution is $f(x) = rx$ if and only if*

$$P(z) = (z - r)^m \quad \text{with } m \in \{1, 2\}. \quad (60)$$

Proof. Any such continuous solution is a homeomorphism by Lemma 4.3, and its minimal polynomial divides P . Theorem 6.1 requires the positive root r . Under (60), the only permitted odd multiplicity is one, so $\mu_f = z - r$.

Conversely, if P has any other root, take that real negative root or that nonreal conjugate pair together with a simple root at r . The square-free construction realizes the resulting divisor of P as a minimal polynomial of degree at least two; the realization is not homogeneous linear. If P has no other root but $m \geq 3$, the construction with $\mu_f = (z - r)^3$ supplies a nonlinear solution. Every solution in this nonunit setting fixes zero, so affine and homogeneous-linear rigidity coincide. $\square$

Corollary 8.2 (Unit circle). *Let P be nonconstant, monic, real, and supported on $|z| = 1$. Let M be the multiplicity of 1 in P .*

- (i) *If $M = 0$, there is no increasing continuous solution on $\mathbb{R}$.*
- (ii) *If $M \geq 1$, every increasing continuous solution is the identity if and only if $M = 1$.*
- (iii) *Among equations with an increasing solution, every increasing continuous solution is affine if and only if either $M = 1$, or $P(z) = (z - 1)^M$ with $M = 2$ or $M = 3$.*

Proof. If $M = 0$, the classification rules out every minimal divisor. If $M = 1$, the only possible minimal divisor is $z - 1$, regardless of the multiplicities of the other roots. When $M \geq 2$, every translation $x \mapsto x + c$ satisfies the equation, because its minimal polynomial for $c \neq 0$ is $(z - 1)^2$. This proves (i) and (ii).

If $P = (z - 1)^M$ with $M = 2$ or 3 , the only nonidentity minimal divisor allowed by the classification is $(z - 1)^2$. To see directly that its solutions are translations, write an orbit as $f^{cn}(x) = x + n(f(x) - x)$. For $x < y$, order preservation for every $n \in \mathbb{Z}$ gives

$$(y - x) + n((f(y) - y) - (f(x) - x)) > 0. \quad (61)$$

The coefficient of n must be zero, so $f(x) - x$ is constant.

If $M \geq 4$, a nonlinear realization of $(z - 1)^4$ is obtained from $h(t) = t + t^3/3$. If $M \geq 2$ and P has another root, adjoining that negative root or nonreal conjugate pair to $(z - 1)^2$ gives an admissible minimal divisor of degree greater than two. Its realization cannot be affine. Indeed, an affine increasing map with slope different from one has that positive slope as a characteristic root, whereas the only positive root here is one; an affine map of slope one is a translation and has minimal degree at most two. $\square$

These criteria make the radius-1 boundary especially clear. A simple positive root at 1 eliminates all other circular modes for increasing solutions. A double root at 1 supplies enough linear drift to allow nonconstant bounded oscillations. The analytic quartic example realizes exactly that first nonreal possibility. [Added 29 September 2026, batch 41: for arbitrary root moduli, affine and homogeneous-linear rigidity are decided by the universal core $\mathcal{R}(P)$ of Part II, Theorem 17.1; on one circle this gives the two corollaries above (Remark 17.2).]

9 What additional regularity does and does not restore

9.1 Differentiability is enough in the square-free nonunit case

Theorem 9.1. *Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be increasing and continuous, and suppose its minimal iterative polynomial is square-free and supported on $|z| = r \neq 1$. If f is differentiable at zero, then $f(x) = rx$ for every x .*

Proof. We already know that $f(0) = 0$. Write $L = f'(0)$. Differentiating the iterative relation at zero, using the chain rule and the fixed point, gives $\mu_f(L) = 0$. Since an increasing differentiable map has $L \geq 0$, and zero is not a root, the only possible value is $L = r$.

For a fixed starting point, square-freeness gives $u(n) = r^n T(n)$, where T is a bounded finite trigonometric sequence. In the time direction in which $u(n) \rightarrow 0$, differentiability implies

$$u(n+1) - ru(n) = o(|u(n)|). \quad (62)$$

Divide by r^{n+1} and use boundedness of T to obtain $T(n+1) - T(n) \rightarrow 0$. This difference is a unit-circle exponential polynomial, so Corollary 5.4 makes it identically zero. In particular, $u(1) = ru(0)$. Since the starting point was arbitrary, $f = r \text{ id}$. $\square$

9.2 A positive-order Taylor remainder handles repeated roots

Theorem 9.2. *Let f be increasing and continuous, with a minimal iterative polynomial supported on $|z| = r \neq 1$, allowing repeated roots. Suppose f is differentiable at zero and, for some $\alpha > 0$,*

$$f(x) = f'(0)x + O(|x|^{1+\alpha}) \quad (x \rightarrow 0). \quad (63)$$

Then $f(x) = rx$ on $\mathbb{R}$. In particular, local $C^{1,\alpha}$ regularity at zero is sufficient.

Proof. As above, $f(0) = 0$ and $f'(0) = r$. Each orbit has the form $u(n) = r^n v(n)$, where v is a unit-circle exponential polynomial and $|v(n)| = O((1+|n|)^d)$ for some d . Apply (63) in the time direction in which $r^n \rightarrow 0$. It gives

$$v(n+1) - v(n) = \frac{u(n+1) - ru(n)}{r^{n+1}} \quad (64)$$

$$= O\left(r^{\alpha n} (1+|n|)^{d(1+\alpha)}\right) \rightarrow 0. \quad (65)$$

The last limit holds both for $r < 1, n \rightarrow +\infty$ and for $r > 1, n \rightarrow -\infty$. By Corollary 5.4, the difference vanishes identically. Thus $u(1) = ru(0)$ for every starting point. $\square$

9.3 C^1 alone is not sufficient with repeated roots

Consider the particularly simple coordinate

$$h(t) = e^t(t^2 + 4), \quad h'(t) = e^t((t+1)^2 + 3) > 0. \quad (66)$$

Use $f = h \circ T \circ h^{-1}$ on $(0, \infty)$ and extend it oddly through zero. Its minimal iterative polynomial is $(z - e)^3$. For $x = h(t)$,

$$\frac{f(x)}{x} = e \frac{(t+1)^2 + 4}{t^2 + 4} \rightarrow e \quad (t \rightarrow -\infty). \quad (67)$$

Also $f'(h(t)) = h'(t+1)/h'(t) \rightarrow e$ as $t \rightarrow -\infty$. Thus the extension is C^1 at zero and consequently C^1 on $\mathbb{R}$. It is nonlinear, because its minimal polynomial has degree three.

For every $\alpha > 0$, however,

$$\frac{|f(h(t)) - eh(t)|}{h(t)^{1+\alpha}} = \frac{e|2t+1|e^{-\alpha t}}{(t^2+4)^{1+\alpha}} \rightarrow \infty \quad (t \rightarrow -\infty). \quad (68)$$

So the positive-order remainder hypothesis of Theorem 9.2 genuinely strengthens bare differentiability. The analytic quartic of Section 3, on the other hand, shows that even real analyticity cannot restore a universal elimination rule at radius one, where the map need not have any fixed point.

10 Exact verification and reproducible artifacts

10.1 What the exact program checks

The script `code/verify.py` requires only Python 3.9 or later and its standard library. All algebraic checks use `Fractions.Fraction`; there is no floating-point arithmetic in these certificates. It implements the cubic function in two independent ways: the direct two-branch scaled formula and the polygonal conjugacy (14).

The recorded run tested 4,026 distinct rational starting points, including positive and negative points, multiple scales, and exact breakpoints. At every point it checked the cubic identity, oddness, scaling, agreement of the two implementations, and inverse identities. It also checked 4,025 adjacent exact secant slopes against the bounds $1/2$ and 4 . These finite tests supplement the all-real branch proof; they are not its logical replacement.

The program separately verifies recurrences on the integer range $-80 \leq n \leq 80$ and computes nonzero exact Hankel determinants for four sample orbits. The symbolic exponential-polynomial arguments explain why the recurrences hold for every integer and why the determinants certify the stated minimal orders.

Table 2: Exact sample certificates. The companion JSON file records the full matrices and polynomial coefficients.

Minimal polynomial	Degree	Hankel determinant
$z^3 - 8$	3	-56
$(z - 1)^2(z^2 + 1)$	4	1
$(z - 2)^3$	3	-512
$(z - 2)^5(z + 2)^4(z^2 + 4)^3$	15	nonzero exact integer

For the degree-15 example the sampled coordinate is

$$h(n) = 2^n \left((n^2 + 64)^2 + \frac{n^2}{1024} \cos \frac{\pi n}{2} + \frac{n^3}{1024} \cos \pi n \right). \quad (69)$$

Every sampled trigonometric value is an integer, so the whole sequence is rational. The real-variable version satisfies the derivative bounds in Section 6.4. Indeed, $k = 2$, $C = 64$, $1/2 < \log 2 < 1$, and $\pi < 4$ give

$$\frac{k}{\sqrt{C}} = \frac{1}{4} < \frac{\log 2}{2}, \quad \frac{2 \log 2 + 5 + 3\pi/2}{1024} < \frac{13}{1024} < \frac{\log 2}{4}. \quad (70)$$

Thus this is not merely a recurrence sequence detached from a monotone function; the construction realizes it as an actual orbit.

10.2 Numerical illustration, clearly separated

The optional script `code/analytic_demo.py` evaluates (17) by bisection. Since $|h(t) - t| \leq 1/4$, the inverse of x is bracketed by $x - 1/4$ and $x + 1/4$ before iteration begins. On 2,001 points in $[-10, 10]$, the recorded double-precision run gave a maximum absolute quartic residual of approximately $7.11 \cdot 10^{-15}$ and a maximum absolute residual in $f^{\circ 4}(x) = x + 4$ of approximately $1.24 \cdot 10^{-14}$. These numbers are implementation checks, not error bounds valid for all inputs and not part of the proof.

The plotting option generates the two vector figures used in this paper. The exact script does not require a plotting library. The article's counterexample remains independently checkable using the three-row itinerary table and the small integer or rational determinants.

11 Conclusions and the remaining boundary

The selected root-elimination question has a definite negative answer under its stated general hypotheses. The self-similar cubic has a real root and a nonreal pair at the same maximal modulus, yet every one of those roots is required by a minimal relation. The analytic quartic shows that this conclusion is not an artifact of a nonsmooth example.

The circular-spectrum classification identifies the mechanism more precisely. Away from radius one, positivity of the normalized two-sided orbit requires an even maximal polynomial degree, producing an odd multiplicity at the positive root. On the unit circle, it is the orbit *increment* that must have constant sign; taking the increment lowers the degree at one, producing an even multiplicity there and a strict multiplicity margin over every oscillating mode. Small trigonometric perturbations realize all and only the permitted patterns.

Several boundaries remain outside this report. We have not classified minimal polynomials with roots on several distinct circles. We have not classified decreasing homeomorphisms, for which consecutive iterates reverse order. Nor have we handled noninjective equations with zero constant coefficient or arbitrary proper interval domains. These are limitations of the proved statements, not qualifications needed by the two counterexamples on $\mathbb{R}$. [Added 29 September 2026, batch 41: Part II classifies minimal polynomials with roots on several distinct circles for increasing homeomorphisms of $\mathbb{R}$ (Theorem 12.2) and gives the exact universal reduction of every equation with nonzero constant coefficient over its increasing solutions (Theorem 16.3). Decreasing homeomorphisms, zero constant coefficient and proper interval domains remain outside both parts.]

Finally, mathematical verification and historical priority are different questions. The paper gives complete arguments and exact finite certificates; it does not supply an independent referee report or an exhaustive priority search. The broader linearity problem is not declared solved. What is established here is the failure of the proposed general nonreal-root deletion and an exact, constructive classification in the increasing single-circle setting. [Added 29 September 2026, batch 41: Part II answers the increasing, nonzero-constant-coefficient version of the linearity problem for arbitrary root moduli (Theorem 17.1); the problem for decreasing maps is still not declared solved.]

Part II. The increasing spectral core on several circles

Sections 1–11 and Appendices A–B form the original report (Part I) of 20 September 2026, unchanged except for a second subtitle line, a paragraph on the title page and dated pointers to this addition. Part II was added on 29 September 2026 in batch 41 of ProveIt’s incoming-report intake, from one manuscript written against the text of Part I printed above. It answers the first boundary that Part I names in Section 11, “We have not classified minimal polynomials with roots on several distinct circles”, for increasing homeomorphisms of $\mathbb{R}$: Theorem 12.2 classifies the minimal iterative polynomials of increasing homeomorphisms with arbitrary root moduli, and Theorem 16.3 gives the exact universal order reduction for every prescribed equation. The other boundaries of that paragraph (decreasing maps, zero constant coefficient, proper intervals) remain open. Section 12 records the provenance and fixes the notation of the addition.

12 Provenance, conventions and the main results

12.1 One manuscript, one addition

Part II is batch-41 manuscript 02, *The Complete Increasing Spectral Core: Arbitrary-radius classification, optimal order reduction, and fixed-point spectra for polynomial iterative equations*, an AI-assisted research draft prepared for Vladimir Reshetnikov and dated 28 September 2026 (20 pages as delivered). It arrived as the archive `ProveIt_Increasing_Spectral_Core.zip` (inner directory `Increasing_Spectral_Core/`) in ProveIt commit 9754e8360 and was placed in eaf787d50. The manuscript is pinned not by a commit but by the Git blob of the file it inspected: Part I’s `article.tex` with blob ae6ea0aa0566316ba9ed06b1750c6e73275059da, retrieved through the GitHub content API on 28 September 2026. That blob is the one committed in dc54c3cb3 and was unchanged at the placement commit, so every statement the manuscript makes about Part I refers to the text printed above. Its code, recorded outputs, status note and source audit are shipped beside Part I’s files with the prefix `02-spectral-core-` (Part I’s unprefix files count as 01); its manuscript, PDF, delivery README and checksum ledger are not shipped and survive in the arrival commit.

The manuscript contributes, in the order printed here: the difference-spectrum identity (Section 13); a complete spectral test for positive bilateral recurrences, with an analytic realization (Section 14); the classification of minimal iterative polynomials of increasing homeomorphisms with arbitrary root moduli (Theorem 12.2, proved in Section 15); the explicit universal annihilator $\mathcal{R}(P)$ (Section 16); an exact affine and homogeneous-linear rigidity criterion (Section 17); criteria for realizations with and without fixed points (Section 18); explicit and uniformly bi-Lipschitz examples (Section 19); its finite checks and their limits (Section 20); ten research questions (Section 21); its conclusion (Section 22); a Hankel-certificate lemma (Section 23); and its source audit (Section 24). Every result, proof, example, remark, limitation and question of the manuscript is printed.

Where the merge had to choose.

- (a) The manuscript reproves, so as not to assume Part I, the lemmas on homeomorphisms, the annihilator ideal, two-sided recurrence decomposition, orbitwise least common multiples, simultaneous returns and positive means, and the translation argument for $(z - 1)^2$. Its proofs are the same arguments as

Part I’s; they are printed once, in Part I, and credited where Part II uses them (Section 13.1, Lemma 14.1, Theorem 17.1). The one-circle paragraph of Proposition 14.2 repeats the parity argument of Theorem 5.3; it is kept, because the proposition covers several circles at once.

- (b) Remarks 15.3 and 17.2 were added in the merge. They show that on one circle Theorem 12.2 is Theorem 6.1 and Theorem 17.1 gives Corollaries 8.1 and 8.2; Remark 15.3 also records a finite comparison of the two classifications made in the write phase.
- (c) Four symbols were renamed to avoid collisions (Section 12.2); no normalization changed. The manuscript’s references to “the ProveIt report” now read “Part I”, and its file names are replaced by the shipped names.
- (d) The manuscript’s appendices (a Hankel lemma and its source audit) are printed as the last two sections of Part II, so that Part I’s appendices keep their letters.

Status (manuscript 02). The classification is for continuous increasing maps $\mathbb{R} \rightarrow \mathbb{R}$ and polynomials with nonzero constant coefficient. The proofs are self-contained. This is an AI-assisted, unrefereed research draft; it is not Lean- or Rocq-verified, and no declaration in ProveIt formalizes any statement of Part I or Part II. The extension beyond Part I is explicit, but an exhaustive claim of historical priority is not made. Part II does not classify decreasing solutions, arbitrary proper-interval domains, noninjective equations with zero constant coefficient, or all individual maps realizing an admissible spectrum; its code accepts factored spectra only and does not factor arbitrary polynomials; and no finite test in it certifies the infinite or analytic parts of the proofs.

12.2 Notation across the two parts

Part I’s symbols keep their meanings in Part I. Part II introduces the difference spectrum, positive admissibility and two cores, and reuses several letters that Part I employs differently. Table 3 fixes the meanings used from here on; letters not listed (P , Q , μ_f , E , Ann , r) agree with Part I.

12.3 A specific continuation, not another circular example

Part I proves a circular-spectrum classification and counterexamples to unrestricted deletion of nonreal characteristic roots. Its conclusion expressly leaves spectra on several distinct circles unclassified. Its circular theorem distinguishes a nonunit circle from the unit circle, and its rigidity statements are correspondingly restricted. Part II addresses that stated gap.

The external background is the order-reduction program of Draga and Morawiec [1], including their Problem 6.1 about nonreal roots tied in modulus with a real root. Draga’s subsequent note [2] treats additional elimination cases. Petrov [3] asks for polynomial conditions forcing continuous solutions to be linear. We answer the *increasing, nonzero-constant-coefficient* version of the associated rigidity problem, distinguishing homogeneous linearity from affineness. We do not classify decreasing solutions or every individual solution.

No theorem of Part I is assumed without proof. The classical orbit-recurrence framework and the circular positivity mechanism are reproved in the manuscript, because the passage from them to an arbitrary-radius theorem requires a new sufficiency construction and careful treatment of the root 1; in this report those reproofs are Part I’s arguments and are cited there.

Table 3: Symbols of Part II, with the tempting false reading beside the true one.

Symbol	Meaning in Part II	Not to be confused with
$\mathcal{D}A$	A with <i>one</i> copy of $z - 1$ removed, if present ((72)); the <i>difference spectrum</i>	the dilations D_2, D_8 of Section 2; removal of the whole $(z - 1)$ -primary factor
$\delta(A)$	1 if $A(1) = 0$, else 0	—
$m_A(\lambda)$	multiplicity of the root λ in A	the same notion as Part I's m_λ, m_r, m_1
positive-admissible	Definition 12.1	“realizable”: μ is a minimal polynomial iff $\mu = z - 1$ or $\mathcal{D}\mu$ is positive-admissible (Theorem 12.2)
r_-, r_+	least and greatest root moduli of the polynomial in question (usually $\mathcal{D}\mu$)	Part I's single common radius r
$\mathcal{R}(P)$	the increasing spectral core (Definition 16.1)	the real line $\mathbb{R}$
$\mathcal{K}(B)$	the positive-recurrence core (Section 16)	—
κ	the largest odd integer $\leq m_B(r)$ in (85) (<i>renamed</i> from the manuscript's k)	Part I's k , where $m_r = 2k + 1$
$b_- < b_+$	least and greatest positive roots of B in (86) (<i>renamed</i> from $a < b$)	the logarithmic rates $a < b$ of Lemma 14.3
Λ	the large constant of Lemma 14.3 (<i>renamed</i> from T)	a trigonometric sequence $T(n)$; the translation $T(t) = t + 1$
Φ	the glued fixed-point map in Theorem 18.4 (<i>renamed</i> from G)	the envelope G of Lemma 14.3
A	a monic polynomial (a spectrum), usually $\mathcal{D}\mu$	Part I's polynomial <i>function</i> $A(t)$ in (40), (49)
G, W	two-ended envelope and weight, (76), (77); $W = 2^t + 2^{-t}$ in Theorem 19.1	Part I's perturbed envelope $W(t)$ in (40), (53)
g, h	positive coordinate $g = h'$ and its antiderivative	Part I's h plays the same role (a conjugating coordinate)
F	the conjugated map $F(h(t)) = h(t + 1)$ on $h(\mathbb{R})$; the polygonal map of Section 19	Part I's cubic counterexample F (Theorem 2.1)
H_λ, H	real modes (79) and their sum; H is also a polygonal interpolation in Section 19	Part I's conjugating map H of (14)
$\mu_<, \mu_>$	monic factors of μ from roots of modulus < 1 , > 1	—
$\text{Fix}(f)$	the fixed-point set of f	—

12.4 Conventions

Write $f^{\circ 0} = \text{id}$ and, for a real polynomial P ,

$$P[f](x) = \sum_{j=0}^{\deg P} a_j f^{\circ j}(x), \quad P(z) = \sum_{j=0}^{\deg P} a_j z^j. \quad (71)$$

An increasing map is strictly increasing. A map is globally bi-Lipschitz if there are $0 < c \leq C < \infty$ such that $c|x - y| \leq |f(x) - f(y)| \leq C|x - y|$ for all x, y . When a nonzero polynomial annihilates f , its monic least-degree annihilator is denoted by μ_f .

For a nonzero monic polynomial A with $A(0) \neq 0$, set

$$\delta(A) = \begin{cases} 1, & A(1) = 0, \\ 0, & A(1) \neq 0, \end{cases} \quad \mathcal{D}A = \frac{A}{(z-1)^{\delta(A)}}. \quad (72)$$

Thus $\mathcal{D}$ removes *one copy*, not the entire 1-primary factor. For a root λ , write $m_A(\lambda)$ for its multiplicity in A .

Definition 12.1 (Positive-admissible spectrum). A nonconstant monic real polynomial A with $A(0) \neq 0$ is *positive-admissible* if the following conditions hold. Let r_- and r_+ be the least and greatest moduli of its roots. At each extreme radius $r \in \{r_-, r_+\}$, the positive real number r is a root, and

$$m_A(r) = \max_{|\lambda|=r} m_A(\lambda). \quad (73)$$

If $r_- = r_+$, this multiplicity must also be odd. There is no additional condition at a strictly intermediate radius. The constant polynomial 1 is not positive-admissible.

Theorem 12.2 (Complete increasing-spectrum classification). *Let μ be nonconstant, monic, and real, with $\mu(0) \neq 0$. The following are equivalent:*

- (i) $\mu = \mu_f$ for an increasing homeomorphism $f : \mathbb{R} \rightarrow \mathbb{R}$.
- (ii) Either $\mu = z - 1$, or $\mathcal{D}\mu$ is positive-admissible.
- (iii) $\mu = \mu_f$ for a globally bi-Lipschitz increasing homeomorphism, real analytic outside at most one point.

When $\mu \neq z - 1$ and the extreme root moduli of $\mathcal{D}\mu$ bracket 1, the realization in (iii) is real analytic everywhere and has no fixed point.

The phrase “at each extreme radius” is essential. When two or more circles occur, even multiplicities at both extremes are allowed. For example, $(z - 2)^2(z - 4)^2$ is realizable, although $(z - 2)^2$ alone is not. The root 1 must be processed before deciding how many circles remain: $(z - 1)(z - 2)^2$ is not realizable as a minimal polynomial.

12.5 Optimal universal reduction

For any real P with $P(0) \neq 0$, we will construct a monic divisor $\mathcal{R}(P)$ with the exact universal property

$$[P[f] = 0 \implies Q[f] = 0 \text{ for every continuous increasing } f : \mathbb{R} \rightarrow \mathbb{R}] \iff \mathcal{R}(P) \mid Q. \quad (74)$$

If increasing solutions exist, a single globally bi-Lipschitz map with minimal polynomial $\mathcal{R}(P)$ witnesses the sharpness of every retained root and multiplicity simultaneously. Section 16 gives the explicit formula; no search through all polynomial divisors is necessary.

12.6 What is added and what is inherited

Question	Part I	Part II
Minimal spectra	One modulus circle	Arbitrary finite sets of circles
Multiplicity restrictions	Circular parity and dominance	Extreme-circle rules and unrestricted interior multiplicities
Universal reduction	Circular rigidity tests	Exact smallest common annihilator for every P
Realizations	Circular coordinates	Two-ended analytic coordinates and global bi-Lipschitz extension
Fixed points	Selected regularity results	Separate spectral existence criteria with and without fixed points

13 From a nonlinear map to a finite-dimensional recurrence

13.1 What is taken from Part I

The manuscript opens by reproving four facts that Part I proves by the same arguments; they are used here in Part I's form.

- (1) A continuous solution $f : \mathbb{R} \rightarrow \mathbb{R}$ of $P[f] = 0$ with $P(0) \neq 0$ is a strictly monotone homeomorphism (Lemma 4.3).
- (2) $\text{Ann}(f)$ is an ideal of $\mathbb{R}[z]$, generated by μ_f when it is nonzero, because $(zP)[f](x) = P[f](f(x))$ uses *right composition* (Lemma 4.2, (25)). One must not distribute f over a linear combination of its values.
- (3) Every bilateral sequence annihilated by $A(E)$, $A(0) \neq 0$, has a unique exponential-polynomial expression $u(n) = \sum_{\lambda} p_{\lambda}(n)\lambda^n$ with $\deg p_{\lambda} < m_A(\lambda)$, and its minimal recurrence polynomial is $\prod_{p_{\lambda} \neq 0} (z - \lambda)^{1 + \deg p_{\lambda}}$ (Lemma 4.4).
- (4) μ_f is the least common multiple of the minimal recurrence polynomials of the orbits $u_x(n) = f^{\circ n}(x)$, $n \in \mathbb{Z}$, and finitely many orbits attain every multiplicity (Lemma 4.5).

The manuscript adds two observations. For a homeomorphism, a factor z could be cancelled from an annihilator by surjectivity; consequently $\mu_f(0) \neq 0$. And if E acts on functions by $E\nu = \nu \circ f$, then on the finite-dimensional cyclic space generated by id it is an invertible linear operator whose cyclic-vector minimal polynomial is μ_f (compare the remark after Lemma 4.5); the same letter E denotes the shift $n \mapsto n + 1$ on sequences, because evaluation intertwines the two actions. The nonlinear nature of f as a real map does not interfere with this linear action on functions.

13.2 Why exactly one factor at one disappears

Lemma 13.1 (Difference-spectrum identity). *For a homeomorphism with minimal polynomial μ , the function $f - \text{id}$ has minimal polynomial $\mathcal{D}\mu$ under right composition by f . Equivalently, $\mathcal{D}\mu$ is the least common multiple of the minimal recurrence polynomials of*

$$d_x(n) = f^{\circ n+1}(x) - f^{\circ n}(x). \quad (75)$$

If $f = \text{id}$, the difference function is zero and its monic annihilator is the constant polynomial 1.

Proof. The difference function is $(E - 1) \text{id}$. A polynomial Q annihilates it if and only if $\mu \mid (z - 1)Q$. The least monic such Q is $\mu/\gcd(\mu, z - 1)$, which is precisely $\mathcal{D}\mu$. Evaluation at every point gives the scalar formulation. $\square$

For increasing f , each nonzero sequence d_x has one strict sign for all $n \in \mathbb{Z}$. Apply $f^{\circ n}$ to the strict inequality $f(x) > x$ or $f(x) < x$. This sign property is the bridge between dynamics and positive recurrence spectra. (Part I uses the same increments (36) on the unit circle only.)

14 Positive bilateral recurrences: a complete spectral test

14.1 The elementary oscillation argument

A finite trigonometric sequence is a finite sum $T(n) = \sum_{\zeta \in S} c_\zeta \zeta^n$ with distinct $|\zeta| = 1$. Two elementary facts will be used.

Lemma 14.1. *There exist integers $N_k \rightarrow +\infty$ such that $\zeta^{N_k} \rightarrow 1$ for every $\zeta \in S$. If a real trigonometric sequence satisfies $\liminf_{n \rightarrow +\infty} T(n) \geq 0$, then $T(n) \geq 0$ for all $n \in \mathbb{Z}$. If such a sequence is nonzero and nonnegative, its constant-frequency coefficient is strictly positive.*

Proof. The first assertion, and the returns $T(N_k + \ell) \rightarrow T(\ell)$ and $T(-N_k + \ell) \rightarrow T(\ell)$ in both time directions, are Lemma 5.1; the second assertion follows from them exactly as in the proof of Theorem 5.3; the third is Lemma 5.2. The manuscript proves all three by the same compactness and Cesàro-mean arguments. $\square$

Proposition 14.2 (Necessary extreme-circle conditions). *The minimal polynomial of a nonzero nonnegative real bilateral recurrence sequence is positive-admissible.*

Proof. Let r_+ be the greatest active root modulus and d_+ the greatest polynomial degree on that circle. As $n \rightarrow +\infty$,

$$u(n) = r_+^n n^{d_+} (T_+(n) + o(1)),$$

where T_+ is the nonzero trigonometric sequence made from the leading coefficients on that circle. Positivity and Lemma 14.1 show that $T_+ \geq 0$ and its constant frequency is positive. Thus the root r_+ occurs with degree d_+ , giving maximal multiplicity.

At $n \rightarrow -\infty$, the least active modulus r_- dominates instead. Apply the same argument to $u(-m)$ as $m \rightarrow +\infty$. The positive root r_- must again attain the greatest multiplicity on its circle.

If there is only one circle, let d be its maximum degree and T its leading trigonometric sequence. Positive time gives $T(\ell) \geq 0$; negative time, using the same simultaneous returns, gives $(-1)^d T(\ell) \geq 0$ for all integers ℓ . Since $T \neq 0$, d is even. The positive-root multiplicity $d + 1$ is therefore odd. (This paragraph is the argument of Theorem 5.3.) $\square$

The two distinct extreme circles use different leading coefficients, so there is no parity relation between them. A parity condition imposed separately on every extreme circle would be false.

14.2 A positive analytic interpolation with any allowed spectrum

Lemma 14.3 (Two-ended positive envelope). *Let $a < b$ and let d, e be arbitrary nonnegative integers. For sufficiently large $\Lambda > 0$, the function*

$$G(t) = e^{at}(\Lambda - t)^d + e^{bt}(\Lambda + t)^e \quad (76)$$

is strictly positive on $\mathbb{R}$ and satisfies

$$cW(t) \leq G(t) \leq CW(t), \quad W(t) = e^{at}(1 + |t|)^d + e^{bt}(1 + |t|)^e, \quad (77)$$

for some constants $c, C > 0$.

Proof. On $[-\Lambda, \Lambda]$ both summands are nonnegative and they do not vanish simultaneously. For $t \geq \Lambda$, the second summand is positive. The first can be negative only when d is odd, and then the ratio of its absolute value to the second is bounded by

$$e^{-(b-a)t} t^{d-e}.$$

Choose $\Lambda \geq 1$ large enough that this bound is at most $1/2$ for all $t \geq \Lambda$. At $t \leq -\Lambda$, the first summand is positive, and a potentially negative second summand is controlled by $e^{-(b-a)|t|} |t|^{e-d}$. Enlarge Λ to make that bound at most $1/2$. This proves positivity everywhere. At $-\infty$, $G/W \rightarrow 1$; the same is true at $+\infty$. A strictly positive continuous quotient has positive minimum and finite maximum on the remaining compact interval, giving (77). $\square$

Theorem 14.4 (Positive recurrence realization). *A nonconstant monic real polynomial A with $A(0) \neq 0$ is positive-admissible if and only if it is the minimal polynomial of a strictly positive bilateral recurrence sequence. Moreover, every such A is the minimal shift polynomial of a strictly positive real-analytic exponential polynomial $g : \mathbb{R} \rightarrow (0, \infty)$ such that*

$$0 < c \leq \frac{g(t+1)}{g(t)} \leq C < \infty \quad (t \in \mathbb{R}). \quad (78)$$

Here a shift polynomial annihilates a function of t under $t \mapsto t + 1$.

Proof. Necessity is Proposition 14.2. For sufficiency, first suppose there are at least two root moduli. Put $a = \log r_-$, $b = \log r_+$, $d = m_A(r_-) - 1$, and $e = m_A(r_+) - 1$. Use G from Lemma 14.3 as the two positive-real-root parts.

For every remaining real root or conjugate pair, choose the representative $\lambda = \rho e^{i\theta}$ with $0 \leq \theta \leq \pi$ and insert the real term

$$H_\lambda(t) = e^{(\log \rho)t} t^{m_A(\lambda)-1} \cos(\theta t). \quad (79)$$

For positive real roots $\theta = 0$; for negative ones $\theta = \pi$. On an extreme circle, the degree condition bounds $|H_\lambda|/W$. For a strictly interior radius, exponential separation at both ends bounds the same quotient, regardless of polynomial degree. Thus, with $H = \sum H_\lambda$, there is $M < \infty$ such that $|H| \leq MW$. Choose a nonzero positive ε small enough that $g = G + \varepsilon H \geq cW/2$. All intended leading mode coefficients are nonzero, and no new mode has been introduced.

If there is only one radius r , write $a = \log r$ and $d = m_A(r) - 1$. The admissibility condition makes d even. Start instead with

$$G(t) = e^{at}(1 + t^2)^{d/2}.$$

Each remaining term (79) has polynomial degree at most d , so a sufficiently small nonzero perturbation again stays between two positive multiples of G .

To justify exact minimality under a real shift, write cosine terms as pairs of complex exponentials. On $e^{st} p(t)$, $E - e^s$ lowers polynomial degree by one. Distinct continuous exponents give independent functions. The exponents $s = \log \rho \pm i\theta$ have $e^s = \lambda, \bar{\lambda}$; when $\theta = \pi$ both give the same negative root, but their leading coefficient functions cannot cancel identically. The maximal degree in that primary component is still $m_A(\lambda) - 1$. It follows that the minimal shift polynomial is exactly A .

The weights $W(t+1)/W(t)$ are bounded above and away from zero: $1 + |t+1|$ and $1 + |t|$ differ by factors between $1/2$ and 2 , and the exponential factors acquire fixed multipliers. The same assertion holds for the single-circle weight. Comparability gives (78).

Finally, for each root the leading coefficient of the sampled sequence $g(t_0 + n)$ is a nonzero real-analytic function of t_0 (or a nonzero complex-valued analytic function). Avoid the finitely many zero sets. Then $g(t_0 + n) > 0$ has minimal polynomial A . $\square$

The last generic-sampling step matters for negative roots. An antiderivative can contain $\sin(\pi t)$, which vanishes at every integer although it is not the zero function. Exact minimality must be proved as a function of the continuous coordinate, or at a nonsingular sampling phase, not inferred carelessly from phase zero.

15 Proof of the increasing-spectrum classification

15.1 Necessity for a family of orbit increments

Lemma 15.1. *The least common multiple of finitely many positive-admissible polynomials is positive-admissible.*

Proof. Take any root at an extreme circle of the least common multiple, with its attained maximal multiplicity. In a factor attaining it, this circle is also extreme; that factor supplies the positive root with at least that multiplicity. If the union has only one circle, every factor has that circle, and the maximum of its positive-root multiplicities is the maximum of finitely many odd integers, hence is odd. $\square$

Suppose $f \neq \text{id}$ is increasing with minimal polynomial μ . By Lemma 13.1, $\mathcal{D}\mu$ is the least common multiple of the nonzero increment-orbit minimal polynomials, using finitely many orbits. Multiply each increment sequence by its constant sign to make it positive. Proposition 14.2 and Lemma 15.1 show that $\mathcal{D}\mu$ is positive-admissible. For $f = \text{id}$ the minimal polynomial is $z - 1$. This proves the necessity in Theorem 12.2.

15.2 Integrating the positive coordinate

Assume $A = \mathcal{D}\mu$ is positive-admissible. Construct $g > 0$ as in Theorem 14.4. We choose a real-analytic antiderivative h whose minimal shift polynomial is exactly μ .

For a nonzero continuous exponent s , a canonical antiderivative is

$$\int e^{st} p(t) dt = e^{st} \sum_{j=0}^{\deg p} \frac{(-1)^j p^{(j)}(t)}{s^{j+1}}. \quad (80)$$

It preserves polynomial degree and hence the shift-root multiplicity. For the frequency $s = 0$, ordinary integration increases polynomial degree by one. Apply this term by term, respecting complex conjugation. If 1 is a root of A , the multiplicity at 1 increases by one, as required. If 1 is absent from A but present in μ , add a nonzero constant, say 1. If 1 is absent from μ , add no constant. Thus $h' = g > 0$ and its exact shift polynomial is μ .

Let $(\alpha, \beta) = h(\mathbb{R})$, with infinite endpoints permitted. Define on this interval

$$F(h(t)) = h(t + 1). \quad (81)$$

The analytic inverse function theorem makes F real analytic there, and

$$F'(h(t)) = \frac{g(t + 1)}{g(t)}. \quad (82)$$

Consequently it is bi-Lipschitz with constants uniform on the interval. Also $P[F](h(t)) = P(E)h(t)$ for every polynomial P , proving exact minimality on that interval.

15.3 The range and extension through a finite endpoint

Write r_- and r_+ for the extreme root moduli of A . The envelope asymptotics give the following exhaustive possibilities:

$$h(\mathbb{R}) = \begin{cases} \mathbb{R}, & r_- \leq 1 \leq r_+, \\ (c, \infty), & 1 < r_-, \\ (-\infty, c), & r_+ < 1. \end{cases} \quad (83)$$

Indeed, at an endpoint with exponential rate zero, g has a nondecaying polynomial envelope and is not integrable. A positive rate is integrable only at $-\infty$, and a negative rate only at $+\infty$. The extreme rates are ordered, so both endpoints cannot be finite.

When the range is $\mathbb{R}$, take $f = F$. It has no fixed point because $h(t + 1) > h(t)$. In a half-line case, set $f(c) = c$ and extend by reflection:

$$f(2c - x) = 2c - F(x) \quad (84)$$

for x in the original half-line. The limiting value of F at c is c , since both $h(t)$ and $h(t + 1)$ tend there. This extension is an increasing homeomorphism. Its bi-Lipschitz bounds hold on each side and pass across c by addition. It is analytic away from c .

If $\mu(1) \neq 0$, our choice of primitive gives $c = 0$. If $\mu(1) = 0$, the constant contribution c is annihilated. Thus in both cases $\mu(1)c = 0$. Iterating (84) gives

$$\mu[f](2c - x) = 2c\mu(1) - \mu[F](x) = 0.$$

The value at c satisfies the same equation. Minimality on the original half-line forces minimality globally. This proves sufficiency and all regularity claims of Theorem 12.2.

Remark 15.2 (What is not being claimed about analyticity). The construction is globally real analytic when its coordinate covers $\mathbb{R}$. In a half-line case the reflected map need not be differentiable at c . No complex-entire inverse is used, and no global complex analyticity is asserted. These distinctions avoid an otherwise incorrect regularity upgrade at an accumulating fixed point.

Remark 15.3 (Added in the merge: one circle recovers Part I). When all roots of μ lie on one circle $|z| = r$, Theorem 12.2 is Theorem 6.1. If $r \neq 1$, then $\mathcal{D}\mu = \mu$ has one circle, and positive admissibility says that r is a root of odd multiplicity at least every other multiplicity: condition (i) of Theorem 6.1. If $r = 1$ and $\mu \neq z - 1$, then $\mathcal{D}\mu$ is nonconstant and lies on the unit circle; it is positive-admissible exactly when 1 is a root of $\mathcal{D}\mu$ of odd multiplicity $m_\mu(1) - 1 \geq 1$ that is at least every other multiplicity, that is, when $m_\mu(1)$ is even, at least 2, and exceeds every other multiplicity: condition (ii). The realizations agree in kind as well: by Theorem 18.3 below, for $\mu \neq z - 1$ on one circle a fixed-point-free realization exists exactly when $r = 1$, in accordance with Section 6 (every realization fixes 0 when $r \neq 1$ and has no fixed point when $r = 1$). As a finite check of the implementation, the manuscript's `realizable` test (shipped as `code/02-spectral-core-spectral_core.py`) was compared in the write phase with the conditions of Theorem 6.1 on 638 single-circle spectra: radii 2 and 1; the roots $\pm r$, the pair $\pm ri$ and the pair $r(3 \pm 4i)/5$; the positive root with multiplicity 0 to 4 and the other three with multiplicity 0 to 3, the empty spectrum excluded (2×319 cases). There were no disagreements. The comparison script is not shipped; the grid above specifies it completely.

16 The exact universal spectral core

16.1 A root-by-root formula

Normalize P to be monic, which does not change its solution set. First define a positive-recurrence core $\mathcal{K}(B)$ for any monic real B with $B(0) \neq 0$.

If B has no positive real root, set $\mathcal{K}(B) = 1$. If its only distinct positive root is r , put

$$\kappa = 2 \left\lfloor \frac{m_B(r) - 1}{2} \right\rfloor + 1, \quad \mathcal{K}(B) = \prod_{|\lambda|=r} (z - \lambda)^{\min(m_B(\lambda), \kappa)}. \quad (85)$$

Thus κ is the largest odd integer not exceeding $m_B(r)$.

If there are at least two distinct positive roots, let $b_- < b_+$ be the least and greatest such roots. Define

$$k_B(\lambda) = \begin{cases} 0, & |\lambda| < b_- \text{ or } |\lambda| > b_+, \\ \min(m_B(\lambda), m_B(b_-)), & |\lambda| = b_-, \\ m_B(\lambda), & b_- < |\lambda| < b_+, \\ \min(m_B(\lambda), m_B(b_+)), & |\lambda| = b_+, \end{cases} \quad (86)$$

and set $\mathcal{K}(B) = \prod_{\lambda} (z - \lambda)^{k_B(\lambda)}$. Conjugate roots receive identical multiplicities, so these are real polynomials.

Definition 16.1 (Increasing spectral core). For monic P with $P(0) \neq 0$, define

$$\mathcal{R}(P) = (z - 1)^{\delta(P)} \mathcal{K}(\mathcal{D}P). \quad (87)$$

The value $\mathcal{R}(P) = 1$ means that no increasing solution exists.

Lemma 16.2. *Every positive-admissible divisor of B divides $\mathcal{K}(B)$. When $\mathcal{K}(B) \neq 1$, it is itself positive-admissible.*

Proof. For an admissible divisor, its extreme root moduli are positive roots of B . Thus all its roots lie between the least and greatest positive roots of B . At a boundary of that full interval, its multiplicities cannot exceed

the positive-root multiplicity there. This gives all bounds in (86). If B has just one positive root, the divisor has only that radius, and its positive-root multiplicity is odd. Formula (85) is exactly the largest permitted cap. Conversely the displayed cores satisfy the defining extreme and parity conditions directly. $\square$

Theorem 16.3 (Sharp universal order reduction). *For every real P with $P(0) \neq 0$, the polynomial $\mathcal{R}(P)$ satisfies:*

- (i) $\mathcal{R}(P) \mid P$, and for continuous increasing maps on $\mathbb{R}$, $P[f] = 0$ if and only if $\mathcal{R}(P)[f] = 0$.
- (ii) Increasing solutions exist if and only if $\mathcal{R}(P) \neq 1$; equivalently, if and only if P has a positive real root.
- (iii) When solutions exist, there is a globally bi-Lipschitz increasing map with minimal polynomial exactly $\mathcal{R}(P)$.
- (iv) For every real polynomial Q , the universal implication (74) holds exactly when $\mathcal{R}(P) \mid Q$.

Proof. Divisibility in (i) follows from the construction. Suppose $P[f] = 0$ and let $\mu = \mu_f$, so $\mu \mid P$. The operation $\mathcal{D}$ preserves divisibility: at the root 1 it replaces multiplicity m by $\max(m - 1, 0)$, a monotone function of m . If $f \neq \text{id}$, the classification makes $\mathcal{D}\mu$ admissible. Lemma 16.2 gives $\mathcal{D}\mu \mid \mathcal{K}(\mathcal{D}P)$. Restoring the possible one copy of $z - 1$ yields $\mu \mid \mathcal{R}(P)$, hence $\mathcal{R}(P)[f] = 0$. For $f = \text{id}$, the factor $z - 1$ in P and in $\mathcal{R}(P)$ gives the same result. The reverse implication follows because $\mathcal{R}(P)$ divides P .

If $\mathcal{R}(P) \neq 1$ but $\mathcal{K}(\mathcal{D}P) = 1$, then $\mathcal{R}(P) = z - 1$, realized by the identity. Otherwise $\mathcal{D}\mathcal{R}(P) = \mathcal{K}(\mathcal{D}P)$: when P has no root at one, neither does its positive core. Theorem 12.2 gives (iii). If $\mathcal{R}(P) = 1$, (i) would force $\text{id} = 0$, impossible on $\mathbb{R}$. A positive root r supplies the solution $f(x) = rx$, and the formula has no nontrivial value without a positive root. This proves (ii).

If $\mathcal{R}(P) \mid Q$, every solution of the reduced equation satisfies $Q[f] = 0$. Conversely, apply any proposed universal relation to the map in (iii); its minimal polynomial forces $\mathcal{R}(P) \mid Q$. When there are no solutions, the implication is vacuous and 1 divides every Q , so (iv) remains valid. $\square$

Equivalently, the ideal of common iterative relations of the increasing solution class is

$$\bigcap_{\substack{f: P[f]=0 \\ f \text{ continuous increasing}}} \text{Ann}(f) = (\mathcal{R}(P)). \quad (88)$$

The intersection over an empty class is the full polynomial ring. This is stronger than merely exhibiting some eliminable roots: it identifies the exact obstruction to *every* further universal reduction.

16.2 Consequences for boundary and interior roots

When $\mathcal{D}P$ has two distinct positive roots, every root strictly between the smallest and largest positive radii survives at its full multiplicity in the universal core. At a boundary, a nonreal or negative root survives up to the multiplicity of the positive root there. It need not disappear. When only one positive root remains, the additional odd cap is exact. The unit root affects these conclusions only through the preliminary one-step multiplicity reduction, but omitting that step changes answers.

The core is idempotent:

$$\mathcal{R}(\mathcal{R}(P)) = \mathcal{R}(P). \quad (89)$$

It also respects divisibility: $P \mid Q$ implies $\mathcal{R}(P) \mid \mathcal{R}(Q)$. This follows either from the sharp universal property or by observing that realizable divisors of P are realizable divisors of Q .

17 An exact answer to increasing affine rigidity

The distinction between a polynomial being the minimal polynomial of *an affine map* and forcing *all maps* to be affine cannot be ignored. A small piecewise-linear example is decisive.

Theorem 17.1 (Affine and homogeneous-linear rigidity). *Assume that P has at least one increasing continuous solution on $\mathbb{R}$. Then all such solutions are affine if and only if*

$$\mathcal{R}(P) = z - r \quad \text{for some } r > 0, \quad \text{or} \quad \mathcal{R}(P) = (z - 1)^2. \quad (90)$$

All such solutions are homogeneous linear if and only if $\mathcal{R}(P) = z - r$ for some $r > 0$. In that case the unique increasing solution is $f(x) = rx$. When $\mathcal{R}(P) = (z - 1)^2$, the solutions are exactly the translations $f(x) = x + c$.

Proof. A reduced polynomial $z - r$ forces $f = r \text{ id}$. If $(z - 1)^2[f] = 0$, the argument in the proof of Corollary 8.2 (each bilateral orbit is $f^{\circ n}(x) = x + n(f(x) - x)$, and order preservation for every $n \in \mathbb{Z}$ forces the coefficient of n in $(y - x) + n((f(y) - y) - (f(x) - x)) > 0$ to vanish) shows that $f(x) - x$ is constant, and every translation does satisfy the equation. The manuscript gives the same argument.

For necessity, let $R = \mathcal{R}(P)$ and use a realization with $\mu_f = R$. An increasing affine map $ax + b$ has minimal polynomial $z - a$ if $b = 0$, $(z - 1)^2$ if $a = 1, b \neq 0$, and $(z - 1)(z - a)$ if $a \neq 1, b \neq 0$. Thus unless R has one of these forms, the realization is nonaffine. The remaining case $R = (z - 1)(z - r)$ with $r > 0, r \neq 1$, also fails universal affine rigidity. Indeed

$$f(x) = \begin{cases} x, & x \leq 0, \\ rx, & x \geq 0 \end{cases} \quad (91)$$

is increasing and continuous, satisfies $R[f] = 0$, and is not affine. This completes the affine classification. Finally nonzero translations are not homogeneous linear, proving the last assertions. $\square$

The piecewise-linear mechanism is already visible in Bogdanov's comment on Petrov's question [3]; it is not claimed as a new construction. What the present result adds is an exact test for arbitrary input spectra, including multiplicities and nonreal roots, through $\mathcal{R}(P)$. The unrestricted version of Petrov's question includes decreasing maps and is not declared solved here.

Remark 17.2 (Added in the merge: the circular rigidity criteria). For P supported on one circle, Theorem 17.1 gives Corollaries 8.1 and 8.2. If $r \neq 1$ and r is a root, then $\mathcal{R}(P) = \mathcal{K}(P)$ is given by (85), and it equals $z - r$ exactly when $\kappa = 1$ and P has no other root, that is, $P = (z - r)^m$ with $m \in \{1, 2\}$. If $r = 1$ and $M = m_P(1) \geq 1$, then $\mathcal{R}(P) = z - 1$ exactly when $\mathcal{D}P$ has no positive root, that is, $M = 1$; and $\mathcal{R}(P) = (z - 1)^2$ exactly when $\mathcal{K}(\mathcal{D}P) = z - 1$, that is, $P = (z - 1)^M$ with $M - 1 \in \{1, 2\}$. By Theorem 16.3(ii), $M = 0$ (or r not a root) leaves no increasing solution. These are the statements of the two corollaries.

18 Which spectra admit fixed points?

18.1 Finite limits eliminate incompatible modes

Lemma 18.1. *Suppose a real recurrence sequence $u(n)$ has a finite limit as $n \rightarrow +\infty$. Its active roots are either of modulus less than 1, or a simple root at 1. Its 1-component equals the limit. With a finite limit as $n \rightarrow -\infty$, replace “less” by “greater”. A recurrence sequence with finite limits at both ends is constant.*

Proof. A greatest-modulus, greatest-degree normalization, as in Proposition 14.2, shows that a nonzero component with modulus greater than one cannot remain bounded: a nonzero finite trigonometric leading term returns away from zero along a subsequence. Apply this successively to remove the finitely many growing levels. On the unit circle, positive polynomial degree is likewise impossible under boundedness. A degree-zero trigonometric sequence having a limit must be constant: its difference from that limit tends to zero, and the simultaneous returns in Lemma 14.1 make it identically zero. The negative-time claim follows by replacing n by $-n$. The intersection of the two permitted supports is the simple constant mode. $\square$

Proposition 18.2 (The fixed set is an interval). *For an increasing polynomial-iterative homeomorphism, $\text{Fix}(f)$ is empty or a closed interval, with points, rays, and the whole line allowed. In particular, a nontrivial bounded component of $\mathbb{R} \setminus \text{Fix}(f)$ cannot occur.*

Proof. The fixed set is closed. If $a < b$ are fixed, every orbit of a point $x \in [a, b]$ stays in $[a, b]$ for all integer times and is monotone. It has finite limits at both ends. Lemma 18.1 makes the orbit constant, so x is fixed. $\square$

18.2 Fixed-point-free realization

Theorem 18.3. *Let $\mu \neq z - 1$ satisfy Theorem 12.2, and let r_-, r_+ be the extreme root moduli of $\mathcal{D}\mu$. A realization with no fixed point exists if and only if*

$$r_- \leq 1 \leq r_+. \quad (92)$$

When it exists it can be globally real analytic and bi-Lipschitz.

Proof. Sufficiency is the full-line case of (83). For necessity, if all roots of $\mathcal{D}\mu$ have modulus greater than one, every increment sequence is absolutely summable toward $-\infty$. Thus every orbit has a finite backward limit, necessarily fixed by continuity. If all roots have modulus less than one, the forward limit is finite instead. Both contradict the absence of fixed points. $\square$

18.3 Fixed-point realization

For a polynomial μ , let $\mu_<$ and $\mu_>$ be its monic factors formed from roots with modulus respectively less and greater than one. An empty factor is 1.

Theorem 18.4 (Fixed-point spectra). *A nonconstant monic real μ , with $\mu(0) \neq 0$, is the exact minimal polynomial of an increasing homeomorphism having a fixed point if and only if either $\mu = z - 1$, or all the following hold:*

- (i) *Its only possible root on the unit circle is 1, with multiplicity at most one.*
- (ii) *At least one of $\mu_<, \mu_>$ is nonconstant.*
- (iii) *Each nonconstant member of $\{\mu_<, \mu_>\}$ is positive-admissible.*

These realizations can be chosen globally bi-Lipschitz and analytic away from one fixed point.

Proof. Suppose a nonidentity realization has a fixed point. By Proposition 18.2, each nonfixed component is a ray with a finite fixed endpoint c . Every orbit on it tends to c in one time direction: otherwise its finite limit would be another fixed point inside the ray. Lemma 18.1 shows that such an orbit has only roots strictly on one side of the unit circle, together with a possible simple constant mode. Consequently (i) holds globally. For increments, the constant mode disappears. The factors $\mu_<$ and $\mu_>$ are separately least common multiples of positive increment-orbit spectra, and hence are admissible by Lemma 15.1. Nonidentity gives (ii).

Conversely construct, for each nonconstant factor, a half-line map with that exact minimal polynomial and finite endpoint zero, using Section 15. Reflection or reversal of the coordinate moves its domain to either desired half-line. If both factors occur, put one on each side of zero. If only one occurs, use it on both sides. The combined map Φ fixes zero, is globally bi-Lipschitz, is analytic away from zero, and has minimal polynomial $\mu_<\mu_>$. There is no root at one in this product.

If μ includes a simple $z - 1$, translate the map: $f(x) = 1 + \Phi(x - 1)$. Any annihilator of f , evaluated at its fixed point 1, must vanish at one. Subtracting this constant contribution shows that it must also annihilate Φ . Conversely the product of those two coprime factors annihilates f . Its minimal polynomial is therefore $(z - 1)\mu_<\mu_>$, as required. $\square$

Theorem 18.4 is genuinely stronger than the general classification. For example $(z - \frac{1}{2})(z - 2)^2$ is admissible globally, but its exterior factor $(z - 2)^2$ is not. Every realization is therefore fixed-point-free. In contrast, $(z - \frac{1}{2})(z - 2)$ has both fixed-point and fixed-point-free realizations. A spectrum does not in general determine the dynamical type of every solution.

19 Explicit examples and arbitrarily high nonreal multiplicity

19.1 Even multiplicities on two extreme circles

Set, for every $n \in \mathbb{Z}$,

$$v_n = 2^n(2 - n) + 4^n(2 + n), \quad (93)$$

$$u_n = 2^n(4 - n) + 4^n\left(\frac{n}{3} + \frac{2}{9}\right). \quad (94)$$

Direct subtraction gives $u_{n+1} - u_n = v_n$. The value v_n is positive: for $|n| \leq 2$ both summands in (93) are nonnegative and not both zero; for $n > 2$ the positive $4^n(n + 2)$ exceeds $2^n(n - 2)$; and for $n < -2$ the corresponding inequality is reversed between the two bases. In fact the same argument works for all real n .

The sequence u_n increases from 0 to $+\infty$. Let H be its strictly increasing polygonal interpolation, $H(n + s) = (1 - s)u_n + su_{n+1}$ for $0 \leq s \leq 1$. Define $F(H(t)) = H(t + 1)$ on the positive half-line and extend oddly through zero. Each orbit at phase s is a linear combination of two shifted copies of u_n , so

$$((z - 2)^2(z - 4)^2)[F] = 0. \quad (95)$$

The exact determinant

$$\det(u_{i+j})_{0 \leq i, j \leq 3} = \frac{16384}{9} \neq 0 \quad (96)$$

certifies minimal degree four, hence the displayed minimal polynomial. The slopes on consecutive intervals are v_{n+1}/v_n . They have positive finite limits 2 and 4 at the two ends; positivity on the remaining finite set gives global bi-Lipschitz bounds. This rational polygonal example is supplemental to the analytic realization theorem, not a substitute for it.

19.2 An everywhere analytic annular example with rational orbit data

Let

$$h(t) = 2^t - 2^{-t} + \frac{1}{10} \sin \frac{\pi t}{2}, \quad f(h(t)) = h(t+1). \quad (97)$$

Since

$$h'(t) = (\log 2)(2^t + 2^{-t}) + \frac{\pi}{20} \cos \frac{\pi t}{2} \geq 2 \log 2 - \frac{\pi}{20} > 0,$$

the coordinate is an analytic bijection $\mathbb{R} \rightarrow \mathbb{R}$ with nonvanishing derivative. The last strict inequality follows, for example, from $\log 2 > 1/2$ and $\pi < 4$. The resulting f is analytic and fixed-point-free, and its minimal polynomial is

$$\mu_f(z) = (z - \frac{1}{2})(z - 2)(z^2 + 1) = z^4 - \frac{5}{2}z^3 + 2z^2 - \frac{5}{2}z + 1. \quad (98)$$

All terms of $h(n)$ are rational. The first four are 0, 8/5, 15/4, 311/40, and

$$\det(h(i+j))_{0 \leq i,j \leq 3} = \frac{225}{256} \neq 0. \quad (99)$$

Thus even the unit-circle nonreal pair survives without any root at one in the characteristic polynomial. The two real roots lie on opposite sides of the unit circle and supply the two ends of the coordinate. (Compare Part I's analytic quartic, Theorem 3.1, where the pair $\pm i$ needs the double root at one that Theorem 6.1 requires on a single circle.)

19.3 No multiplicity penalty inside the annulus

Theorem 19.1 (Uniformly bi-Lipschitz high-order nonreal modes). *For every integer $m \geq 1$, there is an everywhere real-analytic, fixed-point-free increasing homeomorphism f_m with*

$$\mu_{f_m}(z) = (z - \frac{1}{2})(z - 2)(z^2 + 1)^m, \quad \frac{1}{6} \leq f'_m(x) \leq 6 \quad (x \in \mathbb{R}). \quad (100)$$

In particular, a common global bi-Lipschitz bound places no upper bound on the multiplicity of an interior nonreal characteristic root.

Proof. Write $a = \log 2$, $k = m - 1$, and $\varepsilon = a^k / (2k!)$. Set

$$W(t) = 2^t + 2^{-t}, \quad g_m(t) = W(t) + \varepsilon t^k \cos \frac{\pi t}{2}. \quad (101)$$

The exponential-series inequality $e^{a|t|} \geq (a|t|)^k / k!$ implies

$$\varepsilon |t|^k \leq \frac{1}{2} e^{a|t|} \leq \frac{1}{2} W(t).$$

Hence $W/2 \leq g_m \leq 3W/2$. Take the canonical real antiderivative from (80), with no added constant. It is an increasing analytic bijection $h_m : \mathbb{R} \rightarrow \mathbb{R}$. Its exact shift polynomial is the one in (100): integration preserves degree k in the nonzero frequencies $\pm \pi i/2$ and introduces no zero frequency.

Define $f_m = h_m \circ (t \mapsto t+1) \circ h_m^{-1}$. Because $\frac{1}{2} \leq W(t+1)/W(t) \leq 2$,

$$\frac{1}{6} \leq \frac{g_m(t+1)}{g_m(t)} \leq 6.$$

Equation (82) proves the claimed derivative bounds. $\square$

The proof does not infer minimality from the orbit at $h_m(0)$. For certain parities an integrated oscillatory component can vanish at that sampling phase. The shift-space argument proves exact minimality for the map, and a generic phase supplies a full scalar-orbit certificate.

19.4 A complete order-32 to order-23 reduction

Consider the factored polynomial

$$P = (z - 1)^3(z - \frac{1}{2})^2(z - 2)^4(z + 2)^7(z^2 + 1)^5(z^2 + 9)^3. \quad (102)$$

After removing one copy at one, the least and greatest positive roots are $1/2$ and 2 , of multiplicities 2 and 4 . The roots $\pm i$ are interior and keep multiplicity 5 . The boundary root -2 is capped at 4 , and $\pm 3i$ disappear. Restoring the root at one gives

$$\mathcal{R}(P) = (z - 1)^3(z - \frac{1}{2})^2(z - 2)^4(z + 2)^4(z^2 + 1)^5. \quad (103)$$

The degree falls from 32 to 23 . Every increasing solution satisfies the reduced equation, and a single globally analytic bi-Lipschitz solution has exactly that degree- 23 minimal polynomial. Therefore no further universal factor or multiplicity can be removed.

20 Reproducibility and verification boundaries

20.1 Exact code and the scope of its input

Part II ships a standard-library Python implementation, `code/02-spectral-core-spectral_core.py` (delivered as `code/spectral_core.py`). It accepts a factor-level spectrum of Gaussian-rational roots, with rational real and imaginary parts and integer multiplicities. Conjugation symmetry is validated. Squared moduli are rational, so every comparison is exact.

The implementation does *not* claim to factor an arbitrary input polynomial over the algebraic numbers. The theorem applies to all real coefficient polynomials; the shipped implementation has the narrower, explicit input representation above. For rational or real-algebraic coefficients, exact root isolation and comparisons of algebraic squared moduli would extend the front end without changing the core algorithm.

The manuscript's example below imports the module under its delivered name `spectral_core`. Under the shipped names that import fails; run it in a copy of `code/` with the delivered names restored, as the report's `README.md` describes, with that directory on Python's module search path (for example, `PYTHONPATH=code python`).

```
from fractions import Fraction
from spectral_core import Root, increasing_core

one = Root(1)
two = Root(2)
P = {one: 1, two: 2}
assert increasing_core(P) == {one: 1, two: 1}
```

The module also implements tests for exact realizability, fixed-point realizability, fixed-point-free realizability, and affine rigidity. It rejects zero roots, invalid multiplicities, and asymmetric conjugate spectra rather than silently applying the theorem outside its hypotheses.

20.2 Recorded exact checks

The verifier (`code/02-spectral-core-verify.py`, delivered as `code/verify.py`; the file of that name here is Part I's verifier) produced the following results, recorded in `data/02-spectral-core-exact_che`

cks.json and data/02-spectral-core-verification.txt.

Check	Count or exact result
Spectra in two finite multiplicity grids	1354
Divisors examined against the admissibility criterion	97281
Rational recurrence windows for each example	$-80 \leq n \leq 80$
Exact increment/monotonicity assertions	402
Hankel determinant for (94)	16384/9
Hankel determinant for (97)	225/256
Large-core degree reduction	$32 \rightarrow 23$

For each finite input spectrum, the verifier computes the core formula and independently enumerates all divisors satisfying the theorem’s admissibility test. It verifies equality with their least common multiple, core idempotence, divisibility, and exact realizability of every nontrivial core. This checks the finite combinatorics and implementation of the formula, not the analytic realization proof. (Merge note: the enumeration is independent of the core formula but uses the same `positive_admissible` routine as the rest of the module; the comparison with Part I’s independently stated circular theorem is the one in Remark 15.3.)

For each explicit orbit, rational arithmetic verifies the recurrence through the stated window and computes a nonzero Hankel determinant. The recurrence holds for all integers because of the symbolic exponential-polynomial formulas, not because of the finite window. The determinant certifies that no lower-order recurrence can annihilate that orbit; the all-real constructions connect those orbits to actual maps.

The worked example of Section 19 is also a script, `code/02-spectral-core-example.py` (delivered as `code/example.py`); its recorded output is `data/02-spectral-core-example.txt`. In the write phase (29 September 2026, Python 3.14.4) both scripts were rerun in a copy with the delivered names restored: both passed, and their outputs equal the shipped records apart from Windows line endings.

20.3 Rebuilding

The report’s `README.md` gives the rerun commands for both parts and the build command for this article. No bibliography processor is needed; Part II adds no figure files. The JSON output gives full exact matrices, polynomial coefficients in ascending order, and initial values. The short verification log is also included. The complete universal theorem is not machine-checked by these scripts, and no Lean certification is implied.

21 Further research questions and a formalization program

Q1. Decreasing spectra and failure of single-witness sharpness. Determine the exact minimal spectra and universal relation ideal for decreasing homeomorphisms. Simply replacing “positive root” by “negative root” is insufficient: odd-time order reversal and even-time monotonicity must hold simultaneously. Draga’s theorem [2] shows that certain two-negative-root equations reduce to a disjunction of lower-order equations. In such a class, the least common annihilator can require factors that are not simultaneously realized by any one map. The increasing case’s single-witness theorem may therefore have no direct decreasing counterpart.

Q2. Smoothness at a fixed point with several radii. Classify spectra realizable by globally C^k , $C^{k,\alpha}$, or real-analytic maps having a fixed point. The present construction is analytic off the point, and Part I

(Section 9) establishes some single-circle regularity obstructions. Several radii introduce competing power scales and possible resonances. An exact threshold theorem would refine Theorem 18.4, rather than merely improve a construction.

- Q3. The least possible bi-Lipschitz distortion.** For an admissible μ , minimize a symmetric constant L in $L^{-1}|x-y| \leq |f(x)-f(y)| \leq L|x-y|$ over realizations with minimal polynomial μ . Extreme recurrence growth gives natural lower bounds, while Theorem 19.1 shows that large interior multiplicities need not force large distortion. Determine whether the infimum depends only on extreme radii, and when it is attained.
- Q4. All solutions, not only their possible spectra.** Describe the moduli of increasing solutions of a fixed equation. The realization theorem selects convenient coordinates but is not a parametrization of all solutions. One possible route is a positive cone of recurrence initial data together with an order-compatible choice of a fundamental interval and a gluing law.
- Q5. Convexity, concavity, and stronger order restrictions.** Find the analogue of $\mathcal{R}(P)$ for globally convex or concave increasing solutions. Convexity controls differences between nearby orbits, not just signs of individual increments. It may yield stronger multiplicity or oscillation restrictions than the positive-recurrence criterion.
- Q6. Proper intervals and one-sided iteration.** Classify self-maps of bounded intervals, half-lines, and proper open intervals without surjectivity. The proof here uses bilateral orbits in an essential way. A one-sided recurrence need not have the negative-time parity obstruction, and a finite endpoint changes which modes can occur.
- Q7. A certified algebraic-coefficient front end.** Implement exact factorization and algebraic modulus comparison for $P \in \mathbb{Q}[z]$, returning $\mathcal{R}(P)$ together with root-isolation certificates. Equal-modulus decisions, rather than approximate root clustering, are essential at the sharp boundary. The factor-level code shipped with Part II is a testable starting point, not that completed front end.
- Q8. Stability of the core under coefficient perturbation.** Determine the strata on which the degree and multiplicity pattern of $\mathcal{R}(P)$ are constant, and describe their jumps. Equal-modulus and positive-real-root collisions are natural boundary loci. A useful output would distinguish stable eliminations from eliminations destroyed by an arbitrarily small perturbation.
- Q9. Constructive positive envelopes and rational certificates.** The sufficiency proof uses an existentially large Λ and a small perturbation parameter. Produce explicit certified bounds in terms of root separations, degrees, and coefficient heights. This would turn an existence construction into a predictable numerical and symbolic realization algorithm.
- Q10. Lean formalization in separable stages.** A practical sequence is: the annihilator ideal and difference-spectrum identity; bilateral recurrence decomposition; finite trigonometric returns and the positive-mean lemma; the necessary spectral test; the finite root-level core formula; then the analytic positive-envelope and conjugacy construction. The algebraic reduction and the finite classification can be audited independently before the analytic realization library is complete. (As of 29 September 2026 none of these stages has been started in ProveIt.)

These are proposed projects, not claims of unresolved status throughout all literature. The decreasing and regularity questions require a renewed bibliographic check before being presented as newly formulated problems.

22 Conclusion of Part II

Removing one 1-factor converts the increasing iterative-equation problem into a problem about positive bilateral recurrences. At two different ends, different exponential envelopes can dominate. That permits even extreme multiplicities and arbitrary interior multiplicities, exactly where the circular theorem no longer supplies the answer.

The resulting classification is both necessary and sufficient, and its sufficiency is global: every permitted spectrum is realized by a bi-Lipschitz increasing homeomorphism. The universal core then identifies the exact amount of order reduction available from an arbitrary input polynomial, with one realization witnessing all retained multiplicities. The fixed-point theorems separate two dynamical regimes, and the analytic family shows that high nonreal multiplicity is compatible with uniformly controlled metric distortion.

The result closes the several-circle gap in Part I within the increasing whole-line setting. It does not assert a classification of decreasing maps, noninjective zero-constant equations, or every individual solution. Its mathematical claims are the stated and proved theorems; historical priority and independent formal verification remain separate questions.

23 A finite Hankel certificate for minimality

Suppose a scalar sequence u_n is annihilated by a monic degree- d polynomial. If a polynomial of degree $e < d$ also annihilated it, then there would be a nontrivial linear relation among the first $e + 1$ columns of the Hankel matrix $(u_{i+j})_{0 \leq i, j \leq d-1}$: evaluate that recurrence at $i = 0, \dots, d - 1$. Hence a nonzero determinant proves minimal order d . This does not require numerical conditioning estimates. (Part I uses the same argument for (11) and (22).)

For the analytic example, the exact matrix is

$$\begin{pmatrix} 0 & 8/5 & 15/4 & 311/40 \\ 8/5 & 15/4 & 311/40 & 255/16 \\ 15/4 & 311/40 & 255/16 & 5131/160 \\ 311/40 & 255/16 & 5131/160 & 4095/64 \end{pmatrix},$$

with determinant 225/256. Its recurrence polynomial is (98). The stored JSON contains the analogous exact matrix for the even-multiplicity example.

24 Source and novelty audit of Part II

Repository source. The inspected source is Part I, at `SetTheory/Cardinals/docs/reports/log-concavity-and-unimodality/nonreal-roots-in-iterative-equations/article.tex` in ProveIt (<https://github.com/VladimirReshetnikov/ProveIt>), inspected on 28 September 2026. The GitHub content API returned blob SHA

ae6ea0aa0566316ba9ed06b1750c6e73275059da.

The retrieved conclusion explicitly leaves several-circle spectra unclassified. The manuscript does not reuse the circular counterexamples as its new result, and it does not claim that the repository's unrefereed report is a formal certification.

External sources checked. The primary preprints [1, 2] were read for the formulation and scope of the order-reduction problem. Page 13 of [1], including Problem 6.1, was also visually inspected. Petrov's original question and its displayed comments [3], including I. Bogdanov's, were checked directly. (The manuscript cites arXiv:1503.00570 without a version and arXiv:1604.01287v2; Part I's bibliography entries, used here, give the same works.) Searches for broader iterative-spectrum classifications returned a mixture of relevant and irrelevant results; no exhaustive citation-database review was completed. Thus Part II establishes a concrete extension of Part I, not a certified first-in-literature claim.

Trust boundary. All universal statements in Part II rest on the written proofs. The rational scripts check explicit examples and finite consistency of the root-level rule. No specialized theorem from the repository or cited iterative-equation papers is used as an unproved premise; the argument uses standard real analysis and finite-dimensional linear algebra. No independent referee or proof assistant has checked the complete development. The manuscript's author read the repository only; no files were committed or modified there. The manuscript's own status note and source record are shipped verbatim as `02-spectral-core-STATUS.md` and `02-spectral-core-source_audit.md`.

A Source and status audit

The literature search was performed on 20 September 2026. The primary sources below were used to identify the question and delimit the claims, not to replace the proofs in this article.

The target. Draga–Morawiec [1], Problem 6.1, is present in the retrieved HTML and in the rendered page 13 of arXiv:1503.00570v2. The preprint has 14 pages. Its published bibliographic record is *Aequationes Mathematicae* 90 (2016), 935–950, with DOI 10.1007/s00010-016-0420-4. The source is about preserving the solution equation after reducing the characteristic polynomial, rather than merely preserving the existence of at least one solution.

Subsequent context. Draga's note [2] gives another order-reduction theorem and mentions the unrestricted nonreal-root issue in its final remarks. The later paper *Means of iterates* [4] treats a particular family of iterative mean equations and uses characteristic-root analysis. A 2021 paper [5] studies a class of fourth-order equations under restrictions on the characteristic roots. These sources show continuing work on the subject; none of the retrieved material established priority for the counterexample developed here.

Initial lead. Petrov's question [3] was the initial candidate. Its displayed answer count was zero in the retrieved page. An unanswered question page is not, by itself, proof that no published solution exists elsewhere.

Search limitation. The search included the authors’ names, polynomial-like iterative equations, nonreal-root elimination, equal-modulus roots, and the numbered problem. Some returned results were irrelevant and were not used. The search was not a complete citation-database survey. Thus the report does not attach a first-discovery claim to the elementary construction or to the classification theorem. The complete source identifiers and this status distinction are also recorded in `notes/literature_audit.md`.

B Reproduction, selection, and exclusion record

From the archive’s top-level directory, the following commands reproduce the exact certificates and, with the optional plotting dependency, the numerical illustrations:

```
python code/verify.py
python code/analytic_demo.py --plots
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The supplied figure files allow the PDF to be compiled without running the plotting script. No bibliography processor is required.

The exact verifier normalizes positive rational numbers by multiplication or division by eight, not by taking an approximate logarithm. This avoids wrong branch choices at powers of eight. Its independent conjugacy implementation also uses rational operations. The recurrence polynomials in the JSON certificate are stored with coefficients in *ascending* order, so, for example, $z^3 - 8$ is represented by `[-8, 0, 0, 1]`.

The random area draw was:

Number of listed areas	80
Generator call	<code>secrets.randbelow(80)</code>
Returned zero-based index	17
Selected one-based index	18
Selected area	Functional equations
Redraws	0

An entropy-backed draw is recorded rather than seed-replayed; running a new draw is not expected to reproduce the same index. The full 80-element list is in `data/selection.json`.

The excluded input was the 1,095-line file `manifest(1).tex`, with SHA-256 fingerprint

`729fb6e3aef3f30c7c5284c4296681fba72b6d6af68211f178ad0bc3c5d15b41.`

The file was reviewed as an exclusion list; this report does not certify the correctness or priority of its individual mathematical claims. A suggested new manifest entry is supplied separately so that future selections can exclude the present target as well.

References

- [1] S. Draga and J. Morawiec, *Reducing the polynomial-like iterative equations order and a generalized Zoltán Boros' problem*, *Aequationes Mathematicae* **90** (2016), 935–950. doi:10.1007/s00010-016-0420-4. Preprint: arXiv:1503.00570v2, 1 December 2015. Problem 6.1 is on preprint page 13.
- [2] S. Draga, *A note on the order of polynomial-like iterative equations*, *Commentationes Mathematicae* **56** (2016), 243–249. Preprint title: *A note on the polynomial-like iterative equations order*, arXiv:1604.01287v2, 14 November 2016.
- [3] F. Petrov, *Continuous linear recurrent relations*, MathOverflow question 232493, 1 March 2016. <https://mathoverflow.net/questions/232493/continuous-linear-recurrent-relations>. Retrieved 20 September 2026. [Part II, added 29 September 2026: the question's comments by I. Bogdanov, checked for Part II on 28 September 2026, contain the piecewise-linear mechanism used in Theorem 17.1.]
- [4] S. Draga and J. Morawiec, *Means of iterates*, arXiv:1801.05006v2, 19 February 2018.
- [5] S. Thangroongvongthana, V. Laohakosol, and S. Mavecha, *A class of continuous solutions of a fourth order polynomial-like iterative equation*, *Current Applied Science and Technology* **21**(3) (2021), 495–523. <https://li01.tci-thaijo.org/index.php/cast/article/view/248735>.